Trainability of Quantum Circuit Born Machines

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CERTIFICATE

This is to certify that **Kshitij Durge**, BS-MS (Dual Degree) student in Department of Electrical Engineering and Computer Science, has completed bonafide work on the thesis entitled **'Trainability of Quantum Circuit Born Machines'** under my supervision and guidance.

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ABSTRACT

This thesis studies the trainability of Quantum Circuit Born Machines (QCBMs), an implementation of Variational Quantum Algorithms. QCBMs can be realized as generative models with the primary goal of learning a probability distribution that generate a dataset, be it classical or quantum. Minimizing the difference between two such distributions is challenged by exponential concentration and barren plateaus encountered due to an uninformed choice of the parameterized quantum circuit or cost function in the VQA training loop. In a practical scenario, not many cost functions are readily adequate for the job and hence, we conduct a rigorous study of one such metric: the Sinkhorn Diveregence and how it affects the trainability and accuracy of the QCBM. We also propose a novel Dynamic Entorpic Annealing approach for training the QCBM using the Sinkhorn Divergence to improve accuracy while retaining trainability.

LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviations

DEA	Dynamic Entropic Annealing
EWMA	Exponentially Weighted Moving Average
GA	Genetic Algorithm
GAN	Generative Adversarial Network
HEA	Hardware-Efficient Ansatz
HEP	High-Energy Physics
IQP	Instantaneous Quantum Polynomial-time
JSD	Jensen-Shannon Divergence
KL	Kullback-Leibler (Divergence)
LP	Linear Programming
MCMC	Markov Chain Monte Carlo
MMD	Maximum Mean Discrepancy
NISQ	Noisy Intermediate-Scale Quantum
OT	Optimal Transport
QAOA	Quantum Approximate Optimization Algorithm
QCBM	Quantum Circuit Born Machine
QGAN	Quantum Generative Adversarial Network

QML	Quantum Machine Learning
QPU	Quantum Processing Unit
RKHS	Reproducing Kernel Hilbert Space
SGD	Stochastic Gradient Descent
TVD	Total Variation Distance
VQA	Variational Quantum Algorithm
VQE	Variational Quantum Eigensolver

Mathematical Symbols

$\boldsymbol{\theta}$	Vector of classical parameters for the quantum circuit
$U(\boldsymbol{\theta})$	Parameterized unitary operation (the ansatz)
$\|\psi(\boldsymbol{\theta})\rangle$	Parameterized trial quantum state
n	Number of qubits
N	Dimension of the Hilbert space (2^n)
L	Number of repeating layers in the ansatz
M	Number of finite measurement shots
$p(\mathbf{x})$	Target probability distribution
$q_{\boldsymbol{\theta}}(\mathbf{x})$	Model probability distribution generated by the QCBM
$\mathbf{x}, \mathbf{y}$	Classical bitstrings (samples)
$\mathcal{X}$	Discrete state space of all possible bitstrings ($\{0,1\}^n$)
$\mathcal{L}(\boldsymbol{\theta})$	Scalar cost/loss function

$\hat{\mathcal{L}}_t$	Noisy empirical estimate of the loss at epoch t
S_t	Smoothed loss at epoch t (via EWMA filter)
ϵ	Entropic regularization parameter
ϵ_{base}	Initial (base) entropic regularization value
ϵ_{min}	Empirical noise floor (minimum allowed ϵ)
γ	Exponential decay factor for ϵ
$\kappa(\mathbf{x}, \mathbf{y})$	Kernel function (e.g., Gaussian kernel)
σ	Bandwidth parameter of the Gaussian kernel
$c(\mathbf{x}, \mathbf{y})$	Lipschitz ground cost function (e.g., Hamming distance)
$U(\mathbf{x}, \mathbf{y})$	Optimal transport coupling matrix (transport plan)
K	Gibbs kernel matrix ($K = \exp(-c/\epsilon)$)
β	Smoothing coefficient for the EWMA filter
Δ_{min}	Scale-invariant minimum required improvement threshold
τ_{rel}	Relative tolerance factor
τ_{abs}	Base absolute tolerance
P_{limit}	Patience limit (number of epochs before triggering decay)

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Chapter 1

Introduction

1.1 Introduction

Quantum computing represents a fundamental paradigm shift in information processing, promising exponential speedups for specific classes of problems that remain intractable in the classical domain. In the late '90s and early 2000s, breakthroughs such as Shor's algorithm for integer factorization and Grover's algorithm for unstructured search, demonstrated the theoretical superiority and advantage quantum algorithms could provide. However, these algorithms require a lot of physical qubits and fault-tolerant quantum error correction—technologies that, despite rapid acceleration, are still in their infant stage of development

As of 2026, Quantum Computing is witnessing a steady transition from the Noisy Intermediate-Scale Quantum (NISQ) era [1] toward the early stages of quantum utility. While hardware developers have made monumental strides in developing and deploying utility scale processors with several highly connected qubit topologies, full-scale fault-tolerant quantum computing still remains a long term goal. Consequently, executing arbitrarily deep quantum circuits results in a rapid accumulation of errors, necessitating the use of algorithms specifically designed to operate within shallow circuit depths.

To bridge this gap between current hardware limitations and practical quantum advantage, Variational Quantum Algorithms (VQAs) have come to be the leading algorithmic framework [2]. VQAs operate on a hybrid quantum-classical architecture, effectively dividing the computational workload. A quantum processor is utilized to prepare a parameterized quantum state and evaluate a cost function, tasks that operate in the exponentially large Hilbert space [3]. Simultaneously, a classical computer is employed to optimize the parameters of the quantum circuit based on the evaluated cost. By keeping the quantum circuit shallow and offloading the optimization to classical routines, VQAs mitigate the impact of hardware noise while maximizing the utility of state of the art quantum processors.

While VQAs were initially popularized for quantum chemistry via the Variational Quantum Eigensolver (VQE) [4] and combinatorial optimization via the Quantum Approximate Optimization Algorithm (QAOA), their hybrid framework has naturally extended into various other domains. As illustrated in Figure 1.1, VQAs are now actively researched and shown to provide quantum advantage across scientific computing, error mitigation, and the focus of this thesis, Quantum Machine Learning (QML) [5]. Within QML, one of the most viable applications of VQAs is generative modeling. Unlike discriminative tasks that aim to find a single optimal scalar value or decision boundary, generative models aim to solve an unsupervised task to learn and reproduce complex, high-dimensional probability distributions. In this context, Quantum Circuit Born Machines (QCBMs) have been established as a powerful class of implicit generative models [6, 7], operating on the fundamental postulates of the probabilistic nature of quantum mechanics to encode distributions that are classically intractable to sample from [8, 9].

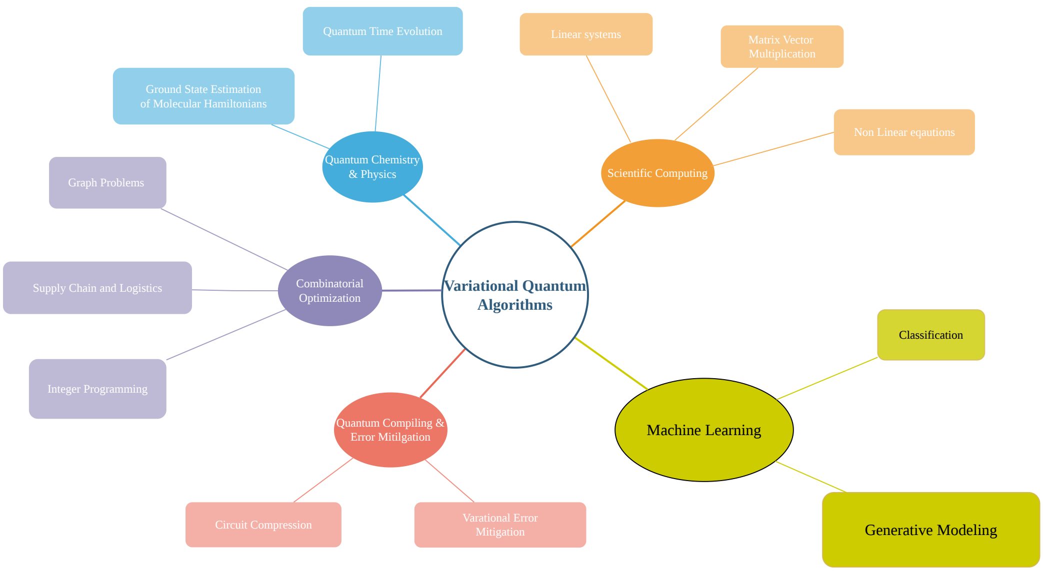

Figure 1.1: Applications of Variational Quantum Algorithms. The hybrid quantum-classical framework has proven versatile, finding applications across domains like quantum chemistry, combinatorial optimization, scientific computing, and quantum machine learning. The focus of this thesis lies in Quantum Generative Modeling

1.2 Trainability Challenges: Barren Plateaus

Despite the theoretical appeal and hardware-friendly nature of VQAs, their scalability is fundamentally challenged by severe trainability issues. As the size of the quantum system scales, the optimization landscape of a VQA can become exponentially flat, a phenomenon known as a *Barren Plateau* (BP) [10]. In a barren plateau landscape, the variance of the gradient of the cost function decays exponentially with the number of qubits, $\mathcal{O}(1/b^n)$ for some $b > 1$. Consequently, navigating this flat landscape requires an exponentially large number of measurement shots to resolve the true gradient signl from statistical noise induced by finite measurements, effectively destroying any potential quantum advantage.

Initial studies attributed barren plateaus to the expressivity of the parameterized quantum circuit. Highly expressive, deep, and unstructured ansatzes naturally form approximate unitary 2-designs, leading to a concentration of measure that flattens the landscape [10, 11]. However, further research revealed that multiple flavors or sources of barren plateaus . As prescribed by Cerezo et al. [12], trainability is heavily influenced not just by the ansatz architecture and hardware noise, but crucially by the mathematical formulation of the cost function itself. For instance, global cost functions, which compare states across the entire Hilbert space, exhibit barren plateaus even for shallow circuits whereas local cost functions can maintain trainability up to logarithmic depths.

1.3 Cost Functions in Quantum Generative Modeling

When working in quantum generative modeling, the choice of cost function becomes even more fragile. The objective of a Quantum Circuit Born Machine (QCBM) is to tune its parameters $\boldsymbol{\theta}$ such that the model distribution $q_{\boldsymbol{\theta}}(\mathbf{x})$ closely approximates a target data distribution $p(\mathbf{x})$. However, QCBMs are fundamentally implicit generative models [7]. This means that while we can efficiently draw samples from the QCBM by measuring the quantum state in the computational basis, we cannot efficiently compute the exact probability amplitudes $q_{\boldsymbol{\theta}}(\mathbf{x})$ for any arbitray computational basis state which are each realized as a certain bitstring $\mathbf{x}$.

The implicitness gives rise to a severe bottleneck when selecting a cost function. Historically, classical machine learning has relied heavily on explicit pairwise loss functions, such as the Kullback-Leibler (KL) Divergence or the Total Variation Distance (TVD), which require direct access to the exact probabilities of both the model and the target distributions. Recent findings by Rudolph et al. [7] show that pairing an implicit quantum model with a pairwise explicit loss function like the KL Divergence leads to a new flavor of barren plateau: exponential loss concentration. Because the empirical samples drawn from the QCBM have an exponentially small probability of overlapping with the exact training dataset in a high-dimensional space, the statistical estimate of the explicit loss concentrates to a fixed value, rendering the model untrainable for large-scale problems.

To address this, the field has pivoted toward implicit loss functions, which only require samples drawn from the distributions rather than exact probabilities. The Maximum Mean Discrepancy (MMD) [13], often realized via a classical Gaussian kernel, is a primary example. By mapping samples into a reproducing kernel Hilbert space, MMD provides a continuous measure of distance between distributions. However, MMD is not the most optimal loss function in terms of trainability and accuracy. Its trainability is highly sensitive to the choice of the kernel bandwidth parameter. As shown in recent literature, an improper bandwidth can cause the MMD observable to become predominantly global, thereby reintroducing a cost-function-induced barren plateau [7]. Furthermore, low-bodied implicit losses often struggle to capture high-order correlations in complex datasets.

1.4 The Finite-Shot Dilemma and Problem Statement

To address such limitations posed by both explicit and standard implicit cost functions, recent theoretical analysis have advocated for Optimal Transport (OT) metrics, specifically the Sinkhorn Divergence, for training QCBMs. Rooted in optimal transport theory [14], the Sinkhorn Divergence introduces an entropic regularization parameter (ϵ) that allows it to interpolate between the computationally intractable Wasserstein distance and the MMD [6].

While the Sinkhorn Divergence theoretically bridges the gap between trainability and accuracy, evaluating this metric on physical, near-term quantum hardware poses another bottleneck. In reality, quantum states must be reconstructed from a strictly limited number of measurements (finite shot regimes). Under these practical constraints, the entropic regularization parameter ϵ presents a bias-variance tradeoff:

- High Regularization (Bias): Setting ϵ to a high value ensures a smooth loss landscape and allows for efficient training with a polynomial number of shots/ measurements. However, it induces severe geometric blurring, permanently limiting the model's accuracy and it is not able to resolve fine details of complex multi-modal target distributions.

- Low Regularization (Variance): Setting ϵ to a low value theoretically captures exact data details. However, the required number of measurement shots scale exponentially for lower regularizations. In finite shot regimes, this causes the gradient signal to be overshadowed by statistical measurement noise, severely degrading the optimization process.

Current methodologies rely on static ϵ regimes, forcing researchers and practitioners to choose between a model that trains efficiently but is highly inaccurate, or a model that is theoretically accurate but impossible to train due to finite shot noise explosions. Furthermore, the exact tolerance of the ϵ parameter to shot noise, even for smaller qubit systems, remains poorly quantified in previous literature. There is a critical need for a framework that can utilize the stable gradients of a highly regularized space while still achieving the exactness of an unregularized metric.

1.5 Thesis Objectives and Contributions

The primary objective of this thesis is to systematically analyze the finite shot limitations of the Sinkhorn Divergence in quantum generative modeling and to propose an autonomous solution to overcome the bias-variance tradeoff.

Specifically, the main contributions of this thesis are twofold:

1. **Empirical Benchmarking of ϵ Tolerance:** We conduct a rigorous, systematic analysis of the behavior of the Sinkhorn loss across various finite shot regimes (from very low shots to infinite shots) and 4 qubit and 8 qubit system. By doing so, we establish the exact threshold where statistical shot noise overshadows the true reliable gradient signal, demonstrating the fundamental failure of stationary regularization strategies.

2. **Dynamic Entropic Annealing (DEA):** We introduce a novel, autonomous training framework designed to navigate the finite-shot dilemma. Rather than relying on a static parameter, DEA treats ϵ as a dynamic hyperparameter. The framework initializes training in a highly regularized (high ϵ) regime to leverage reliable gradient signals and navigate barren plateaus. Utilizing scale-invariant stationarity triggers and optimizer momentum resets, DEA dynamically decays ϵ toward the empirically established noise floor. This allows the QCBM to extract maximal possible accuracy in a single training instance without triggering a noise explosion.

1.6 Thesis Outline

The remainder of this thesis is structured as follows:

Chapter 2: Backgorund and Literature Review provides a survey of the current state of Variational Quantum Algorithms. It traces the discovery and evolution of the barren plateau phenomenon, reviews the development of Quantum Circuit Born Machines, and discusses the intersection of optimal transport theory with quantum machine learning. It details the mechanics of QCBMs, formally defines the barren plateau problem, and rigorously formulates the cost functions evaluated in this study, with a specific focus on the Sinkhorn Divergence and its sample complexity bounds.

Chapter 3: Methodology establishes the mathematical formalisms used in the thesis and outlines the methodology used to obtain our results. It details the choice of parameterized quantum circuits, the target probability distributions, the finite-shot simulation environments, and the algorithmic architecture of the Dynamic Entropic Annealing framework.

Chapter 4: Results and Discussion presents the core findings of our numerical simulations. It first illustrates the limitations of stationary ϵ training by visualizing the bias-variance tradeoff across different shot sizes and demonstrate the empirical noise floor value of ϵ where a shot noise explosion is witnessed. It then demonstrates the superiority of the DEA framework, showing how dynamic ϵ-decay systematically breaks static performance plateaus to reconstruct complex 256-state distributions.

Chapter 5: Conclusion and Future Work summarizes the key insights from this research, discusses the practical limitations of the current study, and proposes future directions for scalable training of quantum generative models in the era of quantum utility.

Chapter 2

Background and Literature Review

This chapter provides a review on the foundations and literature necessary to understand the trainability challenges in quantum generative modeling. We begin by introducing the framework of Variational Quantum Algorithms and their role in the Noisy Intermediate-Scale Quantum (NISQ) era. We then focus on Quantum Circuit Born Machines (QCBMs) as a paradigmatic class of implicit generative models. Subsequently, we explore the fundamental challenges in scaling these models: the barren plateau phenomenon. Finally, we review the evolution of cost functions in quantum machine learning, culminating in the theoretical promise and practical limitations of Optimal Transport metrics, specifically the Sinkhorn divergence, which forms the core of this thesis.

2.1 Variational Quantum Algorithms (VQAs)

The realization of fault-tolerant quantum computation, capable of executing deep algorithms such as Shor's algorithm for prime factorization [15] or Grover's algorithm for unstructured search [16], remains an engineering challenge still. These algorithms require a whole lot pf physical qubits to encode a sufficient number of logical qubits, alongside active quantum error correction protocols that are currently beyond the capabilities of state of the art hardware.

Presently, the field operates within the Noisy Intermediate-Scale Quantum (NISQ) era [1]. NISQ devices are characterized by a limited number of physical qubits that lack error correction and fault tolerance. Consequently, these qubits are highly susceptible to environmental decoherence and operational noise. Executing deep quantum circuits on NISQ hardware results in rapid error accumulation, rendering the final output of the computation indistinguishable from random noise. To extract practical utility from these near-term devices, algorithms must be specifically designed to operate within strict circuit depth constraints.

Variational Quantum Algorithms have emerged as the leading paradigm to address this challenge [2]. VQAs employ a hybrid quantum-classical architecture that divides computational tasks to the processor best suited for them. The quantum processing unit (QPU) is utilized to prepare complex, highly entangled quantum states and evaluate specific observables, operating in the exponentially large Hilbert space. Simultaneously, a classical central processing unit (CPU) is employed to optimize the parameters of the quantum circuit based on the measurement outcomes. By keeping the quantum circuit shallow and assigning the optimization burden to classical routines, VQAs effectively suppress the impact of hardware noise.

2.1.1 The VQA Framework

The execution of a VQA can be formalized as a closed-loop optimization process consisting of three primary components:

1. **Parameterized Quantum Circuit (Ansatz):** The algorithm begins by defining a parameterized unitary circuit, $U(\boldsymbol{\theta})$, known as the *ansatz*. This circuit depends on a set of tunable, and hence optimizable, classical parameters, $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_m)$. When applied to a starting state, typically $|0\rangle^{\otimes n}$, the ansatz prepares a trial quantum state:

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})\,|0\rangle^{\otimes n} \tag{2.1}$$

   The design of the ansatz architecture is critical. It must be sufficiently expressible to explore the desired regions in the Hilbert space containing the optimal solution, yet shallow enough to remain trainable and work with hardware noise [17]. Ansatz architectures generally are categorized into two ways: problem-inspired ansatzes (the Unitary Coupled Cluster ansatz in quantum chemistry) and hardware-efficient ansatzes , which utilize the native gate set and connectivity of the specific quantum device to minimize circuit depth [18].

2. **Cost Function:** The objective of the VQA is encoded into a scalar cost function, $\mathcal{L}(\boldsymbol{\theta})$. This function determines how well the current trial state $|\psi(\boldsymbol{\theta})\rangle$ is able to solve the problem. In many applications, such as the Variational Quantum Eigensolver (VQE) [4], the cost function is defined as the

expectation value of a problem-specific Hamiltonian, $\mathcal{H}$:

$$\mathcal{L}(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | \, \mathcal{H} \, | \psi(\boldsymbol{\theta}) \rangle \tag{2.2}$$

The quantum computer is tasked with preparing the state $|\psi(\boldsymbol{\theta})\rangle$ and performing repeated measurements/ shots to statistically estimate this expectation value.

3. **The Classical Optimizer:** Once the cost function $\mathcal{L}(\boldsymbol{\theta})$ is estimated by the QPU, the value is fed into a classical optimization algorithm. The optimizer updates the parameter vector $\boldsymbol{\theta}$ so as to minimize the cost function:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) \tag{2.3}$$

   This optimization can be performed using gradient-free methods (COBYLA, CMA-ES) or gradient-based methods (Adam, SGD). For gradient-based optimization, the gradients of the quantum circuit can be evaluated analytically on the quantum hardware using techniques such as the parameter-shift rule [19].

The hybrid quantum-classical loop iterates continuously updating parameters, preparing new states, and evaluating the cost, until a convergence criterion is met or a computational budget is exhausted. While this framework is highly versatile, its success is highly dependent on the ability to navigate the cost function landscape, a challenge that becomes particularly intense in the context of quantum machine learning.

2.2 Quantum Generative Modeling and QCBMs

While the VQA framework was initially popularized for solving discriminative and optimization tasks such as finding the ground state energy of a molecule in VQE, its highly flexible architecture is equally suited for unsupervised machine learning. In classical machine learning, tasks are broadly categorized into discriminative and generative models. Discriminative models aim to learn a decision boundary between classes. Generative models, conversely, address a fundamentally harder problem: given a training dataset $\mathcal{D}$ consisting of samples drawn from an unknown target probability distribution $p(\mathbf{x})$ defined over a discrete, high-dimensional space $\mathcal{X} = \{0,1\}^n$, the objective is to construct or to be precise, approximate a parameterized model distribution, $q_{\boldsymbol{\theta}}(\mathbf{x})$, such that $q_{\boldsymbol{\theta}}(\mathbf{x}) \approx p(\mathbf{x})$. A successfully trained generative model can synthesize new, realistic samples that are highly similar the statistical properties of the original data.

2.2.1 The Expressivity vs. Tractability Trade-off

Classical generative modeling is fundamentally constrained by the trade-off between expressivity and computational tractability [20]. This constraint is encountered most when models are categorized into explicit and implicit architectures:

- Explicit Models: These models (Energy-Based Models, Markov Random Fields) allow for the exact computation of the probability $q_{\boldsymbol{\theta}}(\mathbf{x})$ for any given sample. However, to ensure that the probabilities sum to one, these models rely on a normalization constant known as the partition function, $Z = \sum_{\mathbf{x}} e^{-E_{\boldsymbol{\theta}}(\mathbf{x})}$. Computing Z requires summing over all 2^n possible states, a task that is mathematically #P-hard [21]. To make Z tractable, classical explicit models have to restrict their network architectures (assuming variable independence or using directed acyclic graphs). Consequently, they sacrifice expressivity, rendering them incapable of capturing highly complex, deeply correlated real-world data.
- **Implicit Models:** To bypass the partition function bottleneck, models such as Generative Adversarial Networks (GANs) [22] do not rely on the requirement to compute exact probabilities. Instead, they act as black boxes that map latent noise to data samples. Because they do not need to calculate Z, implicit models can utilize massive, unrestricted neural networks, with very high expressivity. However, they sacrifice *tractability*; one can efficiently draw samples from the model, but computing the exact numerical probability of a generated sample is impossible.

2.2.2 The Quantum Circuit Born Machine (QCBM)

To overcome the classical sampling bottlenecks while retaining maximal expressivity, the *Quantum Circuit Born Machine* was introduced as a paradigmatic quantum generative model [13, 23]. While other quantum generative architectures exist theoretically such as Quantum Boltzmann Machines (QBMs) [24], which rely on thermal Gibbs states. They require deep circuits for imaginary time evolution and are practically infeasible on near term hardware. QCBMs, on the contrary, utilize pure quantum states, making them the state f the art for generative modeling in the NISQ era.

QCBMs elegantly resolve the classical expressivity tractability dilemma by utilizing the fundamental postulates of quantum mechanics. First, they obtain normalization for free via unitarity. Because quantum gates are unitary matrices, the state vector is inherently normalized ($\langle\psi|\psi\rangle = 1$), completely rendering the need to compute a classical partition function unnecessary. Second, they achieve massive expressibility/ expressiviry through quantum entanglement and parameterization of Rotation gates, allowing the model to capture high-order correlations between variables that classical neural networks struggle to represent.

A QCBM operates directly within the VQA framework. It utilizes a parameterized quantum circuit, $U(\boldsymbol{\theta})$, acting on an initial state $|0\rangle^{\otimes n}$ to prepare a highly entangled trial pure quantum state $|\psi(\boldsymbol{\theta})\rangle$. According to Born's rule, measuring this quantum state in the computational basis yields a classical bitstring $\mathbf{x} \in \{0,1\}^n$ with a probability equal to the squared magnitude of its probability amplitude:

$$q_{\boldsymbol{\theta}}(\mathbf{x}) = |\langle\mathbf{x}|\psi(\boldsymbol{\theta})\rangle|^2 = \left|\langle\mathbf{x}|\, U(\boldsymbol{\theta})\, |0\rangle^{\otimes n}\right|^2 \tag{2.4}$$

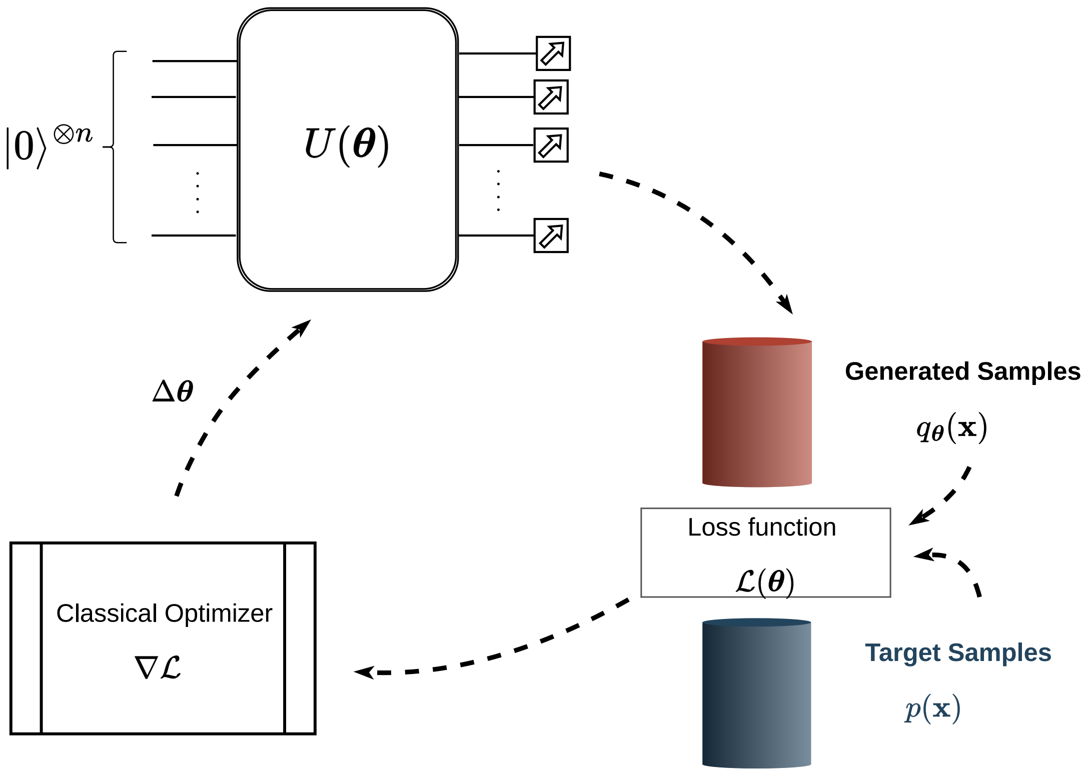

Figure 2.1: The hybrid quantum-classical training loop of a quantum circuit born machine.

As illustrated in Figure 2.1, the training of a QCBM involves repeatedly executing the circuit to gather a batch of generated samples. These samples are compared against the target dataset (samples from the unknown target distribution) using a loss function $\mathcal{L}(\boldsymbol{\theta})$. The classical optimizer then calculates the gradients $\nabla\mathcal{L}$ via the parameter shift rule and updates the circuit parameters $\Delta\boldsymbol{\theta}$ to minimize the distance between the generated and target distributions.

Complexity theory suggests that relatively shallow quantum circuits such as Instantaneous Quantum Polynomial-time (IQP) circuits can generate probability distributions that are strictly impossible to sample from efficiently using any classical randomized algorithm [8, 9]. Therefore, QCBMs are readily able to demonstrate practical quantum advantage.

However, the physical mechanism of the QCBM pose a critical constraint: it is strictly an implicit generative model [25, 7]. Because computing the exact probability amplitude of an arbitrary bitstring in a highly entangled quantum state requires full statevector simulation. This operation scales exponentially $\mathcal{O}(2^n)$ but we only have access to the empirical samples drawn from the physical quantum processing unit. As will be discussed in the following sections, this implicit nature creates a severe bottleneck when attempting to define a trainable cost function, ultimately leading to the barren plateau phenomena that this thesis seeks to resolve.

2.3 Trainability Crisis: Barren Plateaus

While the Variational Quantum Algorithm framework provides a pragmatic pathway to utilizing NISQ devices, its scalability is fundamentally challenged by optimization bottlenecks. The most prominent of these is the Barren Plateau (BP) phenomenon, first formally identified by McClean et al. in 2018 [10]. A barren plateau occurs when the optimization landscape of a parameterized quantum circuit becomes exponentially flat as the system size increases, rendering the model effectively untrainable.

Mathematically, a barren plateau is characterized by a cost function $\mathcal{L}(\boldsymbol{\theta})$ whose gradients vanish exponentially with the number of qubits, n. For a randomly initialized parameter vector $\boldsymbol{\theta}$, the expectation value of the partial derivative with respect to any parameter θ_k is zero:

$$\mathbb{E}_{\boldsymbol{\theta}}\left[\frac{\partial\mathcal{L}(\boldsymbol{\theta})}{\partial\theta_k}\right] = 0 \tag{2.5}$$

More critically, the variance of the gradient decays exponentially:

$$\mathrm{Var}_{\boldsymbol{\theta}}\left[\frac{\partial \mathcal{L}(\boldsymbol{\theta})}{\partial \theta_k}\right] \in \mathcal{O}\left(\frac{1}{b^n}\right) \tag{2.6}$$

for some constant $b > 1$. By Chebyshev's inequality, this implies that the probability of the gradient deviating from zero by any operationally useful amount is exponentially suppressed.

The consequence of a barren plateau on physical quantum hardware is catastrophic. Because quantum expectation values are estimated via finite measurement shots, the empirical gradient is inherently noisy. To resolve an exponentially small gradient signal from the statistical shot noise (which scales as $1/\sqrt{M}$ for M shots), the required number of measurements must scale exponentially, $M \in \mathcal{O}(b^{2n})$ [11, 12]. This completely forbids any potential quantum speedup, as the classical resources required to train the model become intractable.

2.3.1 Expressibility and Unitary 2-Designs

The original and most fundamental source of barren plateaus stems from the expressibility of the parameterized quantum circuit itself. To understand this, one must consider the Haar measure, which represents the translationally invariant probability distribution over the continuous group of unitary matrices, $U(2^n)$. Drawing a unitary from the Haar measure is equivalent to selecting a perfectly uniform, completely random quantum operation, representing maximum expressibility across the entire Hilbert space.

While implementing a truly Haar-random operation requires a circuit of exponential depth, near-term VQAs rely on shallower, parameterized circuits. However, if a parameterized circuit ensemble is sufficiently expressive, it can form an approximate unitary t-design. A unitary t-design is an ensemble of unitaries that perfectly mimics the statistical moments of the Haar measure up to the t-th moment.

Because the variance of the gradient is a second-moment property (involving the tensor product $U \otimes U$), the gradient variance of any circuit that forms a unitary 2-design will be mathematically indistinguishable from the gradient variance of the infinite Haar measure. As proven by McClean et al. [10], the variance of the Haar measure across an n-qubit Hilbert space is $\mathcal{O}(1/2^n)$. Therefore any circuit that forms a 2-design inherits this exponentially vanishing variance, triggering a barren plateau.

This phenomenon creates a severe paradox when utilizing Hardware-Efficient Ansatzes

(HEAs) [18]. HEAs are constructed using alternating layers of parameterized single-qubit rotations and fixed, nearest-neighbor entangling gates. While designed to minimize hardware noise, these alternating layers act as highly efficient mathematical scramblers. Holmes et al. [26] demonstrated a direct correlation between expressibility and trainability. As the depth of an HEA increases (typically requiring only $\mathcal{O}(\text{poly}(n))$ depth), the circuit rapidly converges to an approximate unitary 2-design. Therefore, attempting to make an unstructured ansatz highly expressive inherently destroys its trainability.

2.3.2 Cost-Function-Induced Barren Plateaus

While ansatz expressibility is a critical factor, subsequent research revealed that trainability is equally dictated by the mathematical formulation of the cost function itself. Cerezo et al. [2] demonstrated that even shallow quantum circuits ($\mathcal{O}(1)$ depth), which do not form 2-designs, can exhibit barren plateaus if the cost function is poorly designed.

This discovery established a fundamental dichotomy between *global* and *local* cost functions:

- **Global Cost Functions:** A global cost function measures properties across all n qubits simultaneously. A primary example is the projection onto the all-zero state, $\mathcal{L}_G = 1 - \langle\psi(\boldsymbol{\theta})| \left(|0\rangle\langle 0|\right)^{\otimes n} |\psi(\boldsymbol{\theta})\rangle$. Cerezo et al. proved that global cost functions exhibit barren plateaus for *any* circuit depth, making them inherently unscalable.
- **Local Cost Functions:** A local cost function compares states by measuring single qubits or small, localized neighborhoods of qubits, such as $\mathcal{L}_L = 1 - \frac{1}{n}\sum_{i=1}^{n} \langle\psi(\boldsymbol{\theta})| \left(|0\rangle\langle 0|\right)_i |\psi(\boldsymbol{\theta})\rangle$. Local cost functions were shown to maintain non-vanishing gradients (trainability) up to circuit depths of $\mathcal{O}(\log n)$.

This finding implies that to train a VQA successfully, one must not only utilize shallow circuits but also carefully construct cost functions that avoid global measurements.

2.3.3 Exponential Loss Concentration in Generative Modeling

While the global vs. local dichotomy provides a framework for standard VQAs (like VQE), Quantum Generative Modeling introduces a more severe trainability barrier: *Exponential Loss Concentration* [26, 7].

In generative modeling, the objective is not simply to minimize the energy of a Hamiltonian, but to match a model probability distribution $q_{\boldsymbol{\theta}}(\mathbf{x})$ to a target distribution $p(\mathbf{x})$. Arrasmith et al. established that barren plateaus are intrinsically linked to cost concentration. It is not just the gradient that vanishes. The value of the cost function itself concentrates exponentially tightly around a fixed constant μ:

$$\Pr_{\boldsymbol{\theta}}\left(|\mathcal{L}(\boldsymbol{\theta}) - \mu| \geq \delta\right) \leq \mathcal{O}\left(\frac{1}{b^n}\right) \tag{2.7}$$

Rudolph et al. [7] highlighted how this phenomenon devastates QCBMs. The Hilbert space of an n-qubit system contains 2^n possible bitstrings. When a QCBM is randomly initialized, the probability amplitude is spread thinly across this exponentially large space, meaning the probability of measuring any specific bitstring $\mathbf{x}$ is roughly $1/2^n$.

If a cost function requires exact overlap between the generated samples and the target dataset, the probability of a successful match is exponentially small. Consequently, the cost function evaluates to a fixed, uninformative constant (zero overlap) for almost all parameter configurations, thus destroying reliable gradient computation. The landscape becomes a vast, flat plain with a single, exponentially narrow "gorge" at the global minimum. Because the optimizer only sees the flat plain, it receives no directional gradient information, rendering the generative model untrainable.

This exponential concentration dictates that the choice of cost function in QCBMs is the single most critical factor in achieving scalability, directly motivating the transition from explicit to implicit loss metrics.

2.4 The Cost Function Dilemma in Generative Modeling

The exponential loss concentration phenomenon dictates that the trainability of a Quantum Circuit Born Machine (QCBM) is entirely dependent on the mathematical formulation of its cost function. Because QCBMs are implicit generative models, they provide access only to empirical samples rather than exact probability amplitudes. The choice of cost function must be carefully calibrated to extract meaningful gradient signals from finite measurement shots. Historically, the evolution of cost functions in quantum generative modeling has been characterized by a tension between trainability (mitigating barren pleateaus via reliable gradients) and faithfulness (correctly approximating the target distribution).

2.4.1 The Failure of Explicit Losses

In classical machine learning, generative models are predominantly trained using explicit loss functions. These metrics, such as the Kullback-Leibler Divergence, the Jensen-Shannon Divergence , and the Total Variation Distance, require direct, element-wise comparison between the exact probability of the model $q_{\boldsymbol{\theta}}(\mathbf{x})$ and the target $p(\mathbf{x})$. For example, the KL Divergence is defined as:

$$\mathcal{L}_{\mathrm{KL}}(\boldsymbol{\theta}) = \sum_{\mathbf{x}\in\mathcal{X}} p(\mathbf{x}) \log\left(\frac{p(\mathbf{x})}{q_{\boldsymbol{\theta}}(\mathbf{x})}\right) \tag{2.8}$$

While highly faithful, explicit losses fail catastrophically when applied to implicit quantum models. As rigorously demonstrated by Rudolph et al. [7], evaluating an explicit loss on a QCBM requires constructing a statistical estimate of $q_{\boldsymbol{\theta}}(\mathbf{x})$ from finite measurement shots. However, for a randomly initialized circuit in a high-dimensional Hilbert space, the probability of measuring any specific bitstring present in the training dataset is exponentially small ($\mathcal{O}(1/2^n)$).

Consequently, unless an exponentially large number of shots is employed, the empirical estimate of the overlap between the model and the target distribution is almost always zero. This causes the statistical estimate of the explicit loss to concentrate exponentially tightly around a fixed, uninformative value. The optimizer receives no gradient signal, rendering explicit losses fundamentally untrainable for large-scale quantum generative modeling.

2.4.2 Implicit Losses

To address the issue of exponential concentration of explicit losses, the field transitioned toward implicit loss functions. Implicit losses do not require exact probability amplitudes. Instead they evaluate the distance between distributions by comparing the empirical samples drawn from them.

The most prominent implicit loss in quantum generative modeling is the Maximum Mean Discrepancy (MMD), introduced to the quantum domain by Liu et al. [13] and further analyzed by Coyle et al. [6]. MMD maps the samples from the model and target distributions into a Reproducing Kernel Hilbert Space (RKHS) using a kernel function $\kappa(\mathbf{x}, \mathbf{y})$, which provides a continuous measure of similarity between two bitstrings. The MMD loss is defined as:

$$\mathcal{L}_{\mathrm{MMD}}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{x},\mathbf{y}\sim q_{\boldsymbol{\theta}}}[\kappa(\mathbf{x}, \mathbf{y})] - 2\mathbb{E}_{\mathbf{x}\sim q_{\boldsymbol{\theta}},\mathbf{y}\sim p}[\kappa(\mathbf{x}, \mathbf{y})] + \mathbb{E}_{\mathbf{x},\mathbf{y}\sim p}[\kappa(\mathbf{x}, \mathbf{y})] \tag{2.9}$$

By utilizing a classical Gaussian kernel, $\kappa_G(\mathbf{x}, \mathbf{y}) = \exp(-\|\mathbf{x} - \mathbf{y}\|^2/2\sigma)$, where σ is the bandwidth parameter, MMD bypasses the need for exact bitstring overlap. Even if the generated samples do not perfectly match the target samples, the kernel provides a non-zero distance metric based on their Hamming distance, thereby providing a continuous, trainable gradient signal.

2.4.3 The Weakness and Limitations of MMD

Despite its trainability advantages, MMD suffers from severe theoretical and practical limitations. Coyle et al. [6] established that MMD is a fundamentally weak metric. In probability and statisitcal learning theory, the Total Variation (TV) distance is considered the gold standard for measuring the true distance between distributions. Coyle et al. demonstrated that MMD only provides a *lower bound* on the TV distance:

$$\mathrm{TV}(q_{\boldsymbol{\theta}}, p) \geq \frac{\sqrt{\mathrm{MMD}(q_{\boldsymbol{\theta}}, p)}}{\sqrt{C}} \tag{2.10}$$

where $C := \sup_{\mathbf{x} \in \mathcal{X}^n} \kappa(\mathbf{x}, \mathbf{x})$ represents the supremum (maximum possible value) of the kernel function when comparing any sample $\mathbf{x}$ to itself. Essentially, C bounds the maximum self-similarity score the kernel can assign. For a standard Gaussian kernel, the distance between a point and itself is zero, yielding $\kappa_G(\mathbf{x}, \mathbf{x}) = \exp(0) = 1$. Thus, $C = 1$, making the lower bound immediate. Because MMD only bounds TV from below, minimizing MMD to zero does not mathematically guarantee that the true TV distance is minimized, often leading to inaccurate generative models.

Furthermore, the efficacy of MMD is highly sensitive to the choice of the kernel bandwidth parameter, σ, which introduces a severe trade-off between trainability and faithfulness [7]:

- **Trainability and Globality:** Rudolph et al. showed that the MMD loss can be mathematically mapped to the expectation value of a quantum observable, $\mathcal{O}^{(\sigma)}_{\mathrm{MMD}}$. If the bandwidth σ is chosen to be a constant ($\sigma \in \mathcal{O}(1)$), the effective observable becomes predominantly global, acting non-trivially on $\Theta(n)$ qubits. As established in Section 2.3, global observables induce barren plateaus. Therefore, to maintain trainability, the bandwidth must scale linearly with the number of qubits ($\sigma \in \mathcal{O}(n)$), ensuring the observable remains low-bodied.

- **Faithfulness and Mode Collapse:** While setting $\sigma \in \mathcal{O}(n)$ guarantees trainability, it severely compromises the faithfulness of the loss function. A low-bodied MMD loss is mathematically incapable of distinguishing high-order correlations in the target data. Consequently, the model converges to spurious local minima—learning only the low-order marginals of the distribution while failing to capture the true, complex geometry of the target data. This manifests as mode collapse, where the QCBM fails to reproduce the full diversity of the target distribution.

This inherent tension where explicit losses are faithful but untrainable, and implicit losses like MMD are trainable but weak and prone to mode collapse, gives rise to a critical gap in the literature. It necessitates the exploration of advanced metrics that can provide a strong upper bound on the TV distance while maintaining the trainability of implicit sampling.

2.5 Optimal Transport and the Sinkhorn Divergence

The limitations of the Maximum Mean Discrepancy specifically its status as a weak lower bound on the Total Variation distance and its susceptibility to mode collapse, necessitate the exploration of stronger distance metrics for training Quantum Circuit Born Machines. To guarantee that a generative model accurately captures the target distribution, one requires a loss function that provides a strict upper bound on the TV distance. The mathematical framework of Optimal Transport (OT) provides exactly this capability.

2.5.1 The Optimal Transport Framework

Optimal Transport theory, originally formulated by Monge and relaxed by Kantorovich, provides a rigorous framework for measuring the distance between two probability distributions [14]. Intuitively, it measures the minimum amount of "work" required to transform one distribution (the model $q_{\boldsymbol{\theta}}$) into another (the target p). In the context of discrete probability distributions over bitstrings, this is often referred to as the Earth Mover's Distance or the Wasserstein distance.

Formally, the unregularized Optimal Transport cost between the model distribution $q_{\boldsymbol{\theta}}$ and the target distribution p is defined as:

$$\mathrm{OT}_0^c(q_{\boldsymbol{\theta}}, p) := \min_{U \in \mathcal{U}(q_{\boldsymbol{\theta}}, p)} \sum_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} c(\mathbf{x}, \mathbf{y}) U(\mathbf{x}, \mathbf{y}) \tag{2.11}$$

To understand this formulation, we must define the two core components: the transport plan U and the ground cost c.

Transport Plan (U): The matrix $U(\mathbf{x}, \mathbf{y})$ represents the coupling or transport plan. Each entry in this matrix signifies the exact amount of probability mass that must be moved from state $\mathbf{x}$ in the model distribution to state $\mathbf{y}$ in the target distribution. The set $\mathcal{U}(q_{\boldsymbol{\theta}}, p)$ represents all valid joint probability distributions whose marginals match the model and target exactly. Mathematically, this enforces the conservation of mass: $\sum_{\mathbf{y}} U(\mathbf{x}, \mathbf{y}) = q_{\boldsymbol{\theta}}(\mathbf{x})$ and $\sum_{\mathbf{x}} U(\mathbf{x}, \mathbf{y}) = p(\mathbf{y})$.

Lipschitz Cost Function (c): The function $c(\mathbf{x}, \mathbf{y})$ defines the "ground metric," quantifying the penalty or effort required to move a single unit of mass from $\mathbf{x}$ to $\mathbf{y}$. For quantum generative modeling over n-qubit bitstrings, the natural choice for this cost is the Hamming distance, $d_H(\mathbf{x}, \mathbf{y})$, which counts the number of differing bits between two strings. The Hamming distance is a valid Lipschitz continuous metric, meaning it satisfies the triangle inequality and ensures the geometry of the space is well-behaved. Physically, it reflects the minimum number of bit-flips required to transition from state $\mathbf{x}$ to state $\mathbf{y}$.

The theoretical appeal of Optimal Transport in quantum generative modeling lies in its metric properties. Unlike MMD, the unregularized OT metric (specifically the 1-Wasserstein distance using the Hamming cost) provides a strict upper bound on the Total Variation distance [6]:

$$\mathrm{TV}(q_{\boldsymbol{\theta}}, p) \leq \mathrm{OT}_0^c(q_{\boldsymbol{\theta}}, p) \tag{2.12}$$

Consequently, successfully minimizing the OT cost mathematically guarantees that the TV distance is minimized, ensuring the model faithfully reproduces the target distribution without mode collapse.

2.5.2 Limitations of Unregularized Optimal Transport

Despite its theoretical superiority, unregularized Optimal Transport is practically infeasible for training implicit quantum generative models due to two severe computational bottlenecks:

1. **Computational Complexity of Linear Programming:** Because the objective function $\sum c \cdot U$ is linear with respect to U, and the marginal constraints defining $\mathcal{U}$ are linear equalities, finding the optimal transport plan U^* is fundamentally a Linear Programming problem. For a system of n qubits, the support size is $N = 2^n$, meaning the joint coupling matrix U contains N^2 variables. The computational cost of standard network simplex algorithms used to solve this LP scales as $\mathcal{O}(N^3 \log N)$. This cubic scaling with respect to an already exponentially large Hilbert space makes pure OT classically intractable even for moderate qubit counts.

2. **Intractable Sample Complexity:** Because QCBMs are implicit models, the OT cost must be estimated empirically using M finite measurement shots. However, unregularized OT suffers heavily from the curse of dimensionality. As proven by Dudley [27] and further analyzed in the context of generative models by Genevay et al. [28], the empirical estimation error of unregularized OT scales as $\mathcal{O}(1/M^{1/n})$. To achieve a reliable gradient signal, the required number of measurement shots M must scale exponentially with the number of qubits, rendering the metric unusable on near-term quantum hardware.

2.5.3 Entropic Regularization and the Sinkhorn Divergence

To break the curse of dimensionality and resolve the linear programming bottlenecks of pure OT, Cuturi [29] introduced the concept of entropic regularization. By adding an entropy penalty to the optimal transport objective, the problem is transformed from a computationally expensive, flat-edged linear program into a strictly convex optimization problem.

The entropically regularized Optimal Transport cost is defined as:

$$\mathrm{OT}_\epsilon^c(q_{\boldsymbol{\theta}}, p) := \min_{U \in \mathcal{U}(q_{\boldsymbol{\theta}}, p)} \left(\sum_{\mathbf{x},\mathbf{y}} c(\mathbf{x}, \mathbf{y}) U(\mathbf{x}, \mathbf{y}) + \epsilon \mathrm{KL}(U \parallel q_{\boldsymbol{\theta}} \otimes p) \right) \tag{2.13}$$

where $\epsilon > 0$ is the entropic regularization parameter.

The regularization term utilizes the KL divergence between the transport plan U and the independent tensor product of the marginals, $q_{\boldsymbol{\theta}} \otimes p$. The tensor product $q_{\boldsymbol{\theta}}(\mathbf{x})p(\mathbf{y})$ represents a completely uncorrelated joint distribution—a transport plan where mass is distributed randomly without any geometric preference between the source and target. By penalizing the KL divergence, the regularization forces the optimal transport plan U^* to be fuzzier or more entropic, pushing it toward this independent distribution. This prevents the transport plan from becoming a sparse, deterministic mapping (which causes the LP bottlenecks) and instead forces it to be a dense, smooth matrix.

This strict convexity guarantees a unique optimal coupling matrix and ensures that the gradients with respect to the model parameters $\boldsymbol{\theta}$ are well-behaved and continuous. However, $\mathrm{OT}_\epsilon^c(q_{\boldsymbol{\theta}}, p)$ introduces an entropic bias: the distance between a distribution and itself is no longer zero ($\mathrm{OT}_\epsilon^c(p, p) \neq 0$). To correct this and create a true, symmetric divergence metric, Genevay et al. [28] proposed the **Sinkhorn**

Divergence, which debiases the regularized OT cost:

$$\mathcal{L}^{\epsilon}_{\mathrm{SHD}}(q_{\boldsymbol{\theta}}, p) := \mathrm{OT}^{c}_{\epsilon}(q_{\boldsymbol{\theta}}, p) - \frac{1}{2}\mathrm{OT}^{c}_{\epsilon}(q_{\boldsymbol{\theta}}, q_{\boldsymbol{\theta}}) - \frac{1}{2}\mathrm{OT}^{c}_{\epsilon}(p, p) \tag{2.14}$$

The Sinkhorn Divergence is a highly powerful metric for quantum generative modeling because it smoothly interpolates between two extremes based on the value of ϵ. As $\epsilon \to \infty$, the entropy term dominates, and the Sinkhorn Divergence recovers the Maximum Mean Discrepancy. As $\epsilon \to 0$, the entropy penalty vanishes, and it recovers the pure, unregularized Optimal Transport metric [30].

Most importantly, for $\epsilon > 0$, the Sinkhorn Divergence drastically improves sample complexity. The empirical estimation error drops from the intractable $\mathcal{O}(1/M^{1/n})$ to a highly efficient $\mathcal{O}(1/\sqrt{M})$ [31, 6]. This implies that, unlike pure OT, the Sinkhorn Divergence can be efficiently estimated using a polynomial number of measurement shots, making it a prime candidate for training QCBMs on NISQ devices.

2.5.4 The Sinkhorn-Knopp Algorithm

Beyond its theoretical properties, the entropic regularization introduced in the Sinkhorn Divergence provides a profound computational advantage. Because the regularized loss function is strictly convex, finding the optimal transport plan U^* no longer requires solving a computationally prohibitive Linear Program. Instead, the solution can be found efficiently using matrix scaling via the Sinkhorn-Knopp algorithm [29].

By introducing the Gibbs kernel matrix K, defined element-wise as $K(\mathbf{x}, \mathbf{y}) = \exp\left(-\frac{c(\mathbf{x},\mathbf{y})}{\epsilon}\right)$, the optimal transport plan can be factored into the form $U^* = \mathrm{diag}(\mathbf{u}) K \mathrm{diag}(\mathbf{v})$, where $\mathbf{u}$ and $\mathbf{v}$ are non-negative scaling vectors. The Sinkhorn-Knopp algorithm iteratively updates these vectors to satisfy the marginal constraints of the model ($q_{\boldsymbol{\theta}}$) and the target (p):

$$\mathbf{u}^{(t+1)} = \frac{p}{K\mathbf{v}^{(t)}} \tag{2.15}$$

$$\mathbf{v}^{(t+1)} = \frac{q_{\boldsymbol{\theta}}}{K^T\mathbf{u}^{(t+1)}} \tag{2.16}$$

where the division is performed element-wise. This iterative matrix multiplication is highly parallelizable and converges rapidly. Crucially for Variational Quantum Algorithms, these operations are entirely differentiable, allowing the classical optimizer to seamlessly backpropagate gradients through the Sinkhorn cost function to update the quantum circuit parameters $\boldsymbol{\theta}$.

2.5.5 Sample Complexity and the Epsilon Bounds

While the Sinkhorn Divergence resolves the computational complexity of pure Optimal Transport, its practical utility on quantum hardware is entirely governed by the entropic regularization parameter, ϵ. The choice of ϵ dictates a severe mathematical tension between the sample complexity (efficiency) and the approximation error (accuracy) of the metric.

First, we address the sample complexity. On physical quantum hardware, we do not have access to the exact model distribution $q_{\boldsymbol{\theta}}$; we must estimate the Sinkhorn loss empirically using M finite measurement shots, yielding the approximation $\hat{\mathcal{L}}^{\epsilon}_{\text{SHD}}$. As derived by Genevay et al. [31] and analyzed for QCBMs by Coyle et al. [6], the mean error between the exact Sinkhorn loss and its empirical approximation for an n-qubit system scales as:

$$\mathbb{E}\left|\mathcal{L}^{\epsilon}_{\text{SHD}} - \hat{\mathcal{L}}^{\epsilon}_{\text{SHD}}\right| = \mathcal{O}\left(\frac{1}{\sqrt{M}}\left(1 + e^{\left(\frac{2n^2+n}{\epsilon}\right)}\right)\left(1 + \frac{1}{\epsilon^{\lfloor n/2 \rfloor}}\right)\right) \tag{2.17}$$

To ensure that this loss can be efficiently estimated i.e. meaning the error scales gracefully as $\mathcal{O}(1/\sqrt{M})$ without requiring an exponential number of shots, the exponential terms in the bound must be suppressed. This establishes an **efficiency floor**: ϵ must be chosen such that $\epsilon \geq \mathcal{O}(n^2)$.

Second, we must relate the regularized Sinkhorn cost back to the true Total Variation (TV) distance. The regularized Optimal Transport metric bounds the unregularized Wasserstein metric (and hence the TV distance) via the following inequality:

$$0 \leq \text{OT}^{\ell_1}_{\epsilon}(q_{\boldsymbol{\theta}}, p) - \text{OT}^{\ell_1}_{0}(q_{\boldsymbol{\theta}}, p) \leq 2\epsilon \log\left(\frac{e^2 n}{\epsilon}\right) \tag{2.18}$$

For the regularized OT to provide a strict upper bound on the TV distance, the logarithmic term must remain positive. This establishes an **accuracy ceiling**: ϵ must be chosen such that $\epsilon \leq ne^2$.

Comparing these two bounds reveals a fundamental mathematical contradiction as the system size n grows. If one chooses an ϵ small enough to strictly bound the TV distance ($\epsilon \leq ne^2$), the sample complexity equation picks up an exponential dependence on the dimension n, rendering the loss function efficiently incomputable. Conversely, choosing a large ϵ ($\epsilon \approx \mathcal{O}(n^2)$) guarantees efficient computation but severely weakens the bound on the TV distance, leading to inaccurate generative models.

2.6 Gap in the Literature

The mathematical contradiction between the efficiency floor ($\mathcal{O}(n^2)$) and the accuracy ceiling (ne^2) of the Sinkhorn Divergence presents a severe challenge for near-term implementations. However, as noted by Coyle et al. [6], these bounds represent strict, theoretical worst-case scenarios. The authors conjecture that, in practice, a much lower value of ϵ could likely be chosen without triggering a catastrophic blow-up in sample complexity.

Despite this hypothesis, the existing literature lacks a systematic empirical investigation into the actual tolerance of the Sinkhorn Divergence to shot noise on near-term hardware. When training a QCBM with a limited measurement budget, the static choice of ϵ forces a severe Bias-Variance Tradeoff:

- High ϵ : Operating in a highly regularized regime ensures a smooth, strictly convex loss landscape with reliable gradient signals. However, this induces severe geometric blurring, permanently limiting the model's capacity to resolve fine-grained target distributions.
- Low ϵ : Operating in a weakly regularized regime theoretically captures the exact geometry of the target data. However, pushing ϵ too low inevitably triggers a shot-noise explosion, where the variance of the empirical estimate overwhelms the true gradient signal, stalling the training process entirely.

The exact threshold at which lowering ϵ transitions from improving accuracy to triggering this shot noise explosion remains unmapped, particularly across varying finite measurement budgets.

This unresolved tension motivates the first primary objective of this thesis: a rigorous empirical analysis of stationary ϵ values. By systematically evaluating 8-qubit QCBMs across a wide spectrum of finite shot regimes, we aim to observe the exact practical limits of entropic regularization. As will be detailed in subsequent chapters, our empirical benchmarking confirms the conjecture of Coyle et al., demonstrating that successful training can indeed be achieved at ϵ values significantly lower than the theoretical $\mathcal{O}(n^2)$ bound and in many cases, well below $\epsilon = 1$—before sample complexity explodes.

Nevertheless, while empirical tuning can identify a sweet spot for a static ϵ, any stationary parameterization inherently forces a compromise. A static ϵ low enough to achieve high accuracy often lacks the gradient stability required to navigate the initial barren plateau, while a static ϵ high enough to guarantee initial convergence permanently blurs the final distribution.

To transcend this fundamental limitation, we introduce **Dynamic Entropic Annealing (DEA)**. Rather than settling for a static compromise, DEA autonomously adapts the regularization parameter during a single training instance. By initializing in a highly regularized regime to leverage reliable gradient signals, and subsequently decaying ϵ toward the empirically observed noise floor, DEA extracts maximal accuracy without succumbing to gradient drowning. This dynamic approach systematically outperforms even the most finely-tuned, lowest-viable stationary ϵ baselines, providing a robust and scalable framework for quantum generative modeling in the NISQ era.

Chapter 3

Methodology

3.1 Introduction and Motivation

As established in the theoretical framework, the entropic regularization parameter (ϵ) in the Sinkhorn Divergence dictates a severe mathematical tension between sample complexity and approximation accuracy. Theoretical bounds suggest that to maintain an efficient sample complexity of $\mathcal{O}(1/\sqrt{M})$, the regularization parameter must be restricted to an efficiency floor of $\epsilon \geq \mathcal{O}(n^2)$. Conversely, to strictly bound the Total Variation distance, ϵ must be minimized.

However, as conjectured by Coyle et al. [6], these bounds represent strict, worst-case analytical scenarios. In practical implementations, it is highly probable that a much lower value of ϵ could be utilized without triggering a catastrophic blow-up in sample complexity. Despite this theoretical optimism, there is a distinct lack of empirical testing in the existing literature regarding the actual tolerance of the Sinkhorn Divergence to shot noise on near-term hardware. The exact threshold at which lowering ϵ transitions from improving geometric accuracy to triggering a shot-noise explosion remains entirely unmapped.

This critical gap in the literature serves as the primary motivation for the experimental design of this thesis. To systematically address this, our methodology is divided into two distinct phases.

Phase I consists of a rigorous empirical analysis of stationary ϵ values. We systematically sweep the regularization parameter from the theoretical efficiency floor of $\mathcal{O}(n^2)$ down to the highly unregularized regime of $\mathcal{O}(1)$ and below. By evaluating these stationary values across a wide spectrum of finite measurement budgets, we aim to empirically observe the exact point of failure. This is the specific value of ϵ where the gradient signal is irreversibly drowned by statistical shot noise.

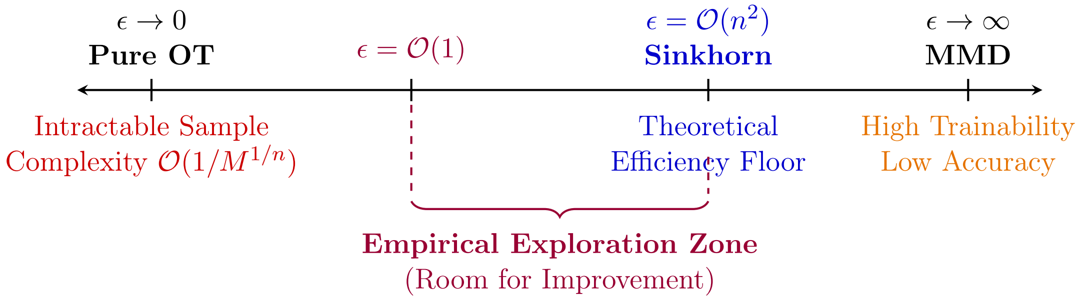

Figure 3.1: The interpolation spectrum of the Sinkhorn Divergence. While theoretical bounds specify an efficiency floor at $\mathcal{O}(n^2)$, this paper empirically explores the intermediate zone down to $\mathcal{O}(1)$ to identify the practical limits of sample complexity.

As illustrated in Figure 3.1, this Empirical Exploration Zone represents the room for improvement over current static methodologies. By mapping this zone, we establish the necessary baselines to justify our ultimate solution.

Phase II introduces the core algorithmic contribution of this work: the **Dynamic Entropic Annealing (DEA)** framework. Informed by the empirical limits discovered in Phase I, DEA abandons the stationary ϵ paradigm entirely. Instead, it autonomously navigates the spectrum shown in Figure 3.1 during a single training instance—initializing safely at the theoretical efficiency floor to leverage reliable gradients, and dynamically decaying into the exploration zone to extract maximal accuracy without triggering a noise explosion.

The subsequent sections of this chapter detail the specific quantum circuit architectures, target distributions, and algorithmic mechanics utilized to execute these two phases.

3.2 Quantum Circuit Architecture (The Ansatz)

The performance of a Quantum Circuit Born Machine (QCBM) is fundamentally dictated by the architecture of its parameterized quantum circuit, commonly referred to as the *ansatz*. The ansatz defines the manifold of quantum states that the model can explore. A successful generative model requires an ansatz that is highly expressive i.e. capable of spanning a vast region of the Hilbert space to capture complex target distributions, while remaining sufficiently shallow to mitigate hardware noise and avoid expressibility-induced barren plateaus.

3.2.1 General Ansatz Formalism

In this thesis, we employ layered Parameterized Quantum Circuits (PQCs). The overall unitary operation $U(\boldsymbol{\theta})$ acting on an n-qubit system is constructed by applying a sequence of L repeating circuit blocks, or layers:

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})\,|0\rangle^{\otimes n} = \left(\prod_{l=1}^{L} W_l(\boldsymbol{\theta}_l)\right)|0\rangle^{\otimes n} \tag{3.1}$$

where $\boldsymbol{\theta}_l$ represents the subset of classical parameters for the l-th layer, and the total parameter vector is $\boldsymbol{\theta} = \bigcup_{l=1}^{L} \boldsymbol{\theta}_l$.

Each layer $W_l(\boldsymbol{\theta}_l)$ generally consists of two primary components:

1. **Single-Qubit Rotations:** Parameterized gates (e.g., $R_x(\theta), R_y(\theta), R_z(\theta)$) applied to individual qubits to explore the local state space.
2. **Entangling Operations:** Two-qubit gates (e.g., CNOT, CZ, or parameterized controlled-rotations like $CR_x(\theta)$) applied across specific qubit pairs to generate quantum correlations and spread information across the system.

The specific choice of gates and the topological connectivity of the entangling operations define the unique characteristics of the ansatz.

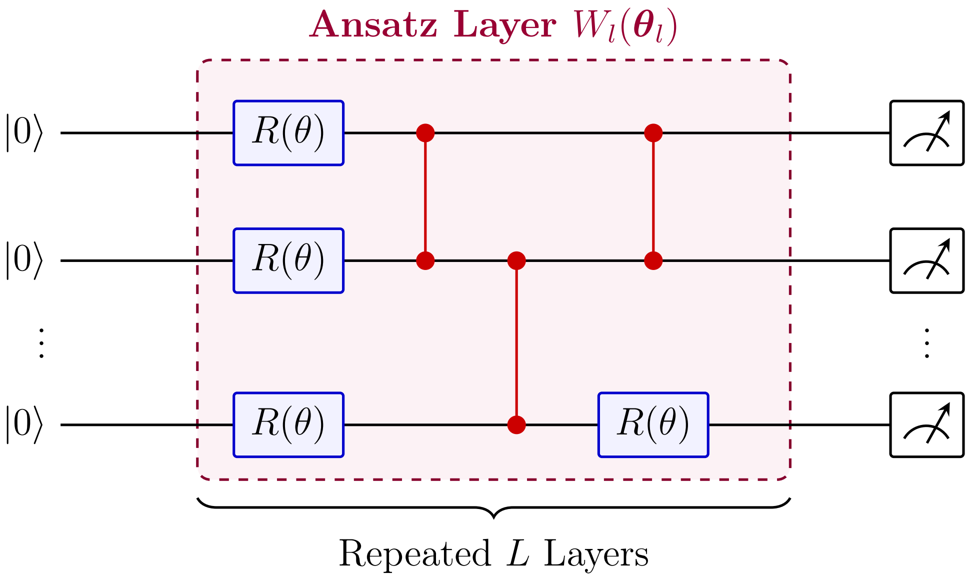

Figure 3.2: Generalized schematic of a layered Parameterized Quantum Circuit for an n-qubit system. An ansatz layer comprises local parameterized rotations and two-qubit entangling operations. As shown, advanced architectures (such as genetically optimized ansatzes) may interleave rotations and entangling gates organically within the same layer, rather than strictly separating them into rigid blocks. The final state is measured in the computational basis to yield empirical samples.

3.2.2 Circuit Categories and Selection

To ensure a robust and comprehensive evaluation of the Sinkhorn Divergence, we benchmark our training frameworks using four distinct circuit architectures. Rather than detailing the exact gate-level composition of each circuit, we categorize them based on their structural paradigms and scaling characteristics. These circuits are drawn from two different design philosophies: standard heuristic templates and algorithmically optimized architectures.

Heuristic Parameterized Templates

Heuristic templates are manually designed, highly structured circuits that rely on repeating uniform blocks of rotations and entangling gates. We utilize two well-established circuit templates from the comprehensive expressibility study by Sim et al. [17]:

- **Circuit A (High Expressibility, Quadratic Cost):** This template utilizes an all-to-all qubit connectivity graph for its entangling operations. While it achieves the highest expressibility scores among the templates evaluated by Sim et al., this comes at a severe resource cost. For an n-qubit system with L layers, the number of parameters, the number of two-qubit gates, and the overall circuit depth all scale quadratically as $\mathcal{O}(n^2L)$. It serves as our highly expressive, parameter-heavy baseline, providing a rigorous test for optimization stability.
- **Circuit B (Moderate Expressibility, Linear Cost):** This template utilizes a circuit-block connectivity, which acts as an intermediate topology between nearest-neighbor and all-to-all interactions. Crucially, it is highly resource-efficient. Its parameter count, two-qubit gate count, and circuit depth all scale linearly as $\mathcal{O}(nL)$. It serves as our baseline for a more hardware-friendly ansatz that maintains moderate expressibility without the quadratic overhead.

Genetically Optimized Ansatzes

While heuristic templates are standard in VQA literature, their rigid, uniform block structures often lead to inefficiencies. To test our methodology on state-of-the-art architectures, we utilize two circuits generated via a Genetic Algorithm (GA) framework, as proposed by Mallapur et al. [32]:

- **Circuit C and Circuit D:** Unlike heuristic templates that strictly separate rotation and entanglement into distinct sub-layers, these genetically evolved circuits interleave single- and two-qubit gates in an unstructured, organic manner. The GA framework explicitly optimizes the gate arrangement to maximize expressibility (by minimizing the Jensen-Shannon divergence to the Haar measure) while strictly constraining the circuit depth to scale linearly, $d \in \mathcal{O}(n)$. These circuits provide highly complex entanglement structures that achieve the expressibility of deep heuristic circuits but at a fraction of the parameter count and depth, representing the cutting edge of resource-efficient ansatz design.

By evaluating our stationary and dynamic training methodologies across these four diverse architectures which span quadratic to linear scaling and are rigid to evolutionary topologies, we ensure that our findings regarding the finite-shot tolerance of the Sinkhorn Divergence are fundamentally robust and not merely artifacts of a specific circuit structure.

3.3 Target Probability Distributions

To evaluate the generative capabilities of the QCBMs and the faithfulness of the Sinkhorn Divergence, we benchmark the models against predefined, classically generated target distributions. For an n-qubit system, the target distribution $p(\mathbf{x})$ is defined over a discrete domain of $N = 2^n$ possible states, where each state corresponds to an integer index $x \in \{0, 1, \ldots, N-1\}$.

In our primary 8-qubit experiments ($N = 256$), we utilize two distinct target distributions to test different aspects of generative performance:

1. **Normal (Gaussian) Distribution:** This serves as a baseline to test the model's ability to converge to a single, continuous global maximum. The unnormalized probability mass function is defined as:

$$\tilde{p}_{\text{normal}}(x) = \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \tag{3.2}$$

   where the mean is centered in the Hilbert space, $\mu = (N-1)/2$, and the standard deviation is set to $\sigma = N/6$ to ensure the distribution tapers off smoothly within the boundaries of the state space.

2. **Bimodal Distribution:** This serves as a significantly harder test case. Generative models trained with weak metrics (such as uncalibrated MMD) frequently suffer from mode collapse, where the model learns the mean of the data but fails to capture distinct peaks. To test the Sinkhorn Divergence's ability to resolve complex geometry, we define a mixture of two Gaussians:

$$\tilde{p}_{\text{bimodal}}(x) = \exp\left(-\frac{1}{2}\left(\frac{x-\mu_1}{\sigma}\right)^2\right) + \exp\left(-\frac{1}{2}\left(\frac{x-\mu_2}{\sigma}\right)^2\right) \tag{3.3}$$

   where the peaks are located at $\mu_1 = N/4$ and $\mu_2 = 3N/4$, with a narrower standard deviation of $\sigma = N/10$.

Both distributions are subsequently L_1-normalized such that $\sum_{x=0}^{N-1} p(x) = 1$ to form valid target probability distributions.

3.4 Phase I: Stationary ϵ Benchmarking

The first phase of our methodology is designed to empirically map the Bias-Variance Tradeoff of the Sinkhorn Divergence and identify the exact threshold of the shot-noise explosion. To achieve this, we execute a massive, highly parallelized grid search over stationary values of the entropic regularization parameter, ϵ.

3.4.1 The Experimental Grid

For each of the four circuit architectures (Circuit 6, Circuit 14, Opt-Circuit 1, and Opt-Circuit 2) and both target distributions, we evaluate the training performance across a two-dimensional grid of hyperparameters:

- **Entropic Regularization (ϵ):** We sweep ϵ from the highly regularized, theoretical efficiency floor down to the ultra-low, unregularized regime. The tested values are $\epsilon \in \{64.0, 48.0, 32.0, 24.0, 16.0, 12.0, 8.0, 4.0, 1.0, 0.5, 0.1, 0.01, 0.005\}$.
- **Measurement Shots (M):** To observe the impact of statistical noise, we evaluate each ϵ value across a spectrum of finite-shot regimes:
  - Very Low: $M \in \{32, 64, 128\}$
  - Moderate (Polynomial): $M \in \{256, 640\}$
  - High: $M = 6400$
  - Infinite: Exact statevector simulation ($M \to \infty$) to serve as the noise-free baseline.

3.4.2 Optimization and Evaluation Metrics

Crucially, for Phase I, we utilize standard **Stochastic Gradient Descent (SGD)** with a fixed learning rate of $\eta = 0.01$. While adaptive optimizers like Adam are standard in VQA literature, they utilize momentum and moving averages of past gradients, which can artificially mask the immediate impact of shot noise. By using vanilla SGD, we ensure that the optimizer's trajectory is a direct, unmasked reflection of the raw gradient variance at that specific ϵ and shot count.

To evaluate the success of the training, we track the exact **Total Variation Distance** between the model's current state and the target distribution at every epoch:

$$\mathrm{TVD}(q_{\boldsymbol{\theta}}, p) = \frac{1}{2} \sum_{x=0}^{N-1} |q_{\boldsymbol{\theta}}(x) - p(x)| \tag{3.4}$$

Because the TVD is tracked analytically (using the exact statevector of the current parameters, independent of the shot noise used to calculate the gradient), it provides an absolute, objective measure of how close the QCBM is to the true target distribution.

By analyzing the late-stage mean and variance of the TVD across this grid, we empirically define the exact boundaries of the Empirical Exploration Zone (as defined in Figure 3.1), setting the stage for our dynamic algorithmic solution.

3.5 Phase II: Dynamic Entropic Annealing (DEA)

The empirical benchmarking in Phase I demonstrates that stationary entropic regularization forces an inescapable compromise: high ϵ guarantees stable convergence but induces geometric blurring (bias), whereas low ϵ captures fine details but triggers a shot-noise explosion (variance). To transcend this limitation, we introduce the **Dynamic Entropic Annealing (DEA)** framework.

DEA abandons the stationary paradigm by treating the regularization parameter ϵ as a dynamic, autonomously decaying hyperparameter. The conceptual goal of DEA is to navigate a shrinking, noisy valley: it initializes training in a highly regularized (wide and smooth) regime to leverage reliable gradient signals, and subsequently decays ϵ (narrowing and sharpening the valley) toward a theoretical noise floor only when the optimizer has extracted all possible geometric information at the current resolution.

3.5.1 Algorithmic Mechanics of DEA

The DEA framework operates as an autonomous wrapper around the standard VQA optimization loop. To successfully navigate the shifting loss landscape, the algorithm must accurately detect when it has reached a local minimum and safely transition to a sharper resolution. This is achieved through four primary mechanical components.

1. Filtering the Noise: The EWMA Filter

In statistical learning theory, we must clearly distinguish between the true mathematical loss (which is inaccessible) and the noisy empirical estimate outputted by the quantum computer. Let $\hat{\mathcal{L}}_t$ denote the raw, noisy empirical Sinkhorn loss evaluated at epoch t.

If the algorithm relied directly on $\hat{\mathcal{L}}_t$ to detect stationarity, the inherent shot-noise jitter would cause the trigger to fire erratically, either prematurely halting exploration or failing to recognize a true plateau. To mitigate this, DEA employs an Exponentially Weighted Moving Average (EWMA) filter. The smoothed loss, S_t, is recursively defined as:

$$S_t = \beta S_{t-1} + (1-\beta)\hat{\mathcal{L}}_t \tag{3.5}$$

where $\beta \in [0, 1)$ is the smoothing coefficient. In our implementations, we utilize $\beta = 0.9$.
Justification: Without the EWMA filter, the algorithm would be blinded by high-frequency statistical noise, making autonomous decision-making impossible in finite-shot regimes.

2. Defining Progress: Scale-Invariant Stationarity

The core decision engine of DEA is its ability to detect when the optimizer has stopped making meaningful progress. This is achieved by tracking the best observed smoothed loss, S_{best}, and comparing it against the current smoothed loss S_t.

However, as ϵ decays, the absolute magnitude of the Sinkhorn loss naturally shrinks. A static tolerance threshold would fail because a significant improvement at $\epsilon = 64.0$ is mathematically much larger than a significant improvement at $\epsilon = 0.5$. Therefore, DEA implements a *scale-invariant* trigger mechanism. The minimum required improvement, Δ_{min}, is defined dynamically:

$$\Delta_{\text{min}} = \max\left(\tau_{\text{rel}}|S_{\text{best}}|, \tau_{\text{abs}}\left(\frac{\epsilon_t}{\epsilon_{\text{base}}}\right)\right) \tag{3.6}$$

This equation relies on two distinct bounds:

- **Relative Tolerance ($\tau_{\text{rel}}|S_{\text{best}}|$):** The algorithm demands an improvement proportional to the current loss magnitude (e.g., $\tau_{\text{rel}} = 0.01$ requires a 1% improvement).
Justification: Without relative tolerance, the trigger would not scale with the shrinking loss values, causing it to fire instantly at lower ϵ regimes where absolute improvements are naturally tiny.
- **Scaled Absolute Tolerance ($\tau_{\text{abs}}(\epsilon_t/\epsilon_{\text{base}})$):** As the loss approaches zero, 1% of nearly zero becomes infinitesimally small, which could cause the algorithm to wait forever for an impossible improvement. To prevent this, we enforce a base absolute tolerance ($\tau_{\text{abs}} = 10^{-5}$) that scales linearly with the current ϵ_t relative to the starting ϵ_{base}.
Justification: Without the scaled absolute floor, the algorithm would suffer

from Zeno's paradox, infinitely chasing microscopic, noise-driven improvements near the global minimum without ever triggering the next decay.

3. Enforcing Patience

Even with a smoothed loss and a scale-invariant threshold, quantum optimization landscapes are highly non-convex and prone to temporary stalls. If the current smoothed loss fails to improve by at least $\Delta_{\min}$ (i.e., $S_t \geq S_{\text{best}} - \Delta_{\min}$), DEA does not immediately drop ϵ. Instead, it increments a patience counter.

Only if the counter exceeds a predefined *Patience Limit* ($P_{\text{limit}} = 75$ epochs) does the algorithm definitively declare stationarity and trigger an annealing event. ***Justification:*** Without a patience limit, the algorithm would prematurely abandon viable optimization paths at the first sign of resistance, failing to fully explore the current ϵ landscape.

4. Optimizer Momentum Wiping

Unlike the stationary benchmarking in Phase I, which utilized standard Stochastic Gradient Descent (SGD) to observe raw gradient variance, the DEA framework employs the **Adam optimizer** [33] to navigate the complex, shifting loss landscape. Adam is an adaptive learning rate optimization algorithm that relies heavily on the exponential moving averages of past gradients (first moment, or momentum) and past squared gradients (second moment, or uncentered variance).

These internal momentum states allow Adam to accelerate convergence along flat ravines and dampen oscillations in steep directions. However, this reliance on historical gradient data introduces a critical vulnerability during dynamic annealing.

When an annealing event occurs ($\epsilon_t \to \epsilon_{t+1}$), the entropic regularization penalty drops instantly. As established by the interpolation properties of the Sinkhorn divergence [30], the geometry of the loss landscape sharpens abruptly; wide, blurred valleys suddenly narrow, and previously hidden local minima emerge. If the optimizer retains its historical momentum from the high-ϵ (blurred) regime, it will apply a massive, outdated update step based on a landscape geometry that no longer exists. This would violently eject the optimizer from the newly revealed, narrower local minimum, destroying the progress made during the previous phase.

To prevent this catastrophic ejection, DEA performs a momentum wipe simultaneously with every ϵ decay. The internal state of the Adam optimizer, specifically the moving average buffers for the first and second moments, is completely reset to zero. This technique draws inspiration from "warm restart" methods in classical machine learning [34], forcing the optimizer to halt its previous trajectory

and explore the newly sharpened landscape using only fresh, localized gradient information evaluated at the new ϵ resolution.
Justification: Without the momentum wipe, the sudden sharpening of the loss landscape would cause the optimizer's outdated momentum to overshoot the true minimum, rendering the dynamic decay of ϵ actively harmful to convergence.

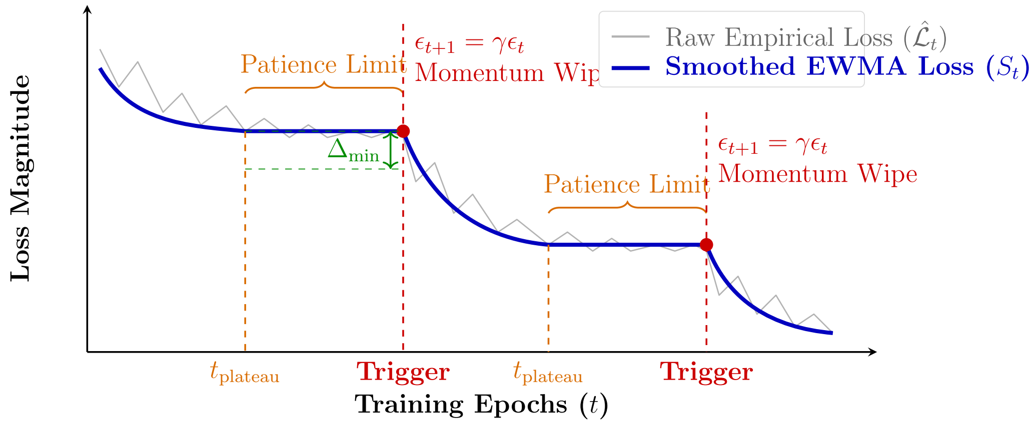

Figure 3.3: Schematic representation of the Dynamic Entropic Annealing (DEA) trigger mechanism. The highly noisy empirical loss ($\hat{\mathcal{L}}_t$, gray) is smoothed via an EWMA filter (S_t, blue). Once the smoothed loss fails to improve by a scale-invariant threshold ($\Delta_{\min}$) for a duration exceeding the Patience Limit, the algorithm autonomously triggers an annealing event. The regularization parameter ϵ is decayed, and the optimizer momentum is wiped. The optimizer then immediately exploits the newly sharpened loss landscape, driving the loss down to a more accurate plateau.

3.6 Software and Simulation Environment

To ensure the reproducibility and computational rigor of the experiments outlined in Phase I and Phase II, all quantum circuit simulations and classical optimization routines were executed within a highly controlled software environment.

The quantum generative models were constructed and simulated using **PennyLane**, an open-source, cross-platform library specifically designed for differentiable programming of quantum computers. The quantum circuits were executed on PennyLane's `default.qubit` statevector simulator. For finite-shot regimes ($M < \infty$), empirical samples were drawn directly from the simulated statevector, and the gradients of the Sinkhorn Divergence were computed using the hardware-compatible *parameter-shift rule* [19]. For the infinite-shot (analytic) baseline, gradients were computed using standard backpropagation via the `autograd` interface to isolate the exact mathematical trajectory of the optimization landscape.

The Sinkhorn-Knopp matrix scaling algorithm, utilized to compute the regularized optimal transport cost, was implemented natively in `numpy`. A critical computational challenge arises in the unregularized regime ($\epsilon \to 0$); as the regularization parameter shrinks, the elements of the Gibbs kernel matrix $K = \exp(-C/\epsilon)$ decay exponentially, leading to severe `float64` underflow. To maintain numerical stability during the iterative updates of the scaling vectors $\mathbf{u}$ and $\mathbf{v}$, a strict lower bound ($\epsilon_{\min} = 0.5$) was enforced during dynamic annealing, and a machine-epsilon safeguard (10^{-15}) was added to the denominator of the scaling equations to prevent division-by-zero errors.

Given the massive scale of the experimental grid—comprising 252 independent stationary training instances in Phase I and 56 dynamic annealing instances in Phase II, with each instance running for up to 1,000 epochs—the computational workload was highly parallelized. The grid search was distributed across a high-performance computing environment utilizing Python's `concurrent.futures.ProcessPoolExecutor`, allowing up to 50 independent training trajectories to be evaluated simultaneously. This parallelized architecture ensured that the extensive empirical mapping of the finite-shot dilemma could be executed efficiently and robustly.

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the empirical evaluation of the Sinkhorn Divergence in quantum generative modeling, systematically addressing the finite-shot dilemma outlined in Chapter 3. The assessment is divided into two primary phases. First, we conduct a rigorous stationary analysis using Stochastic Gradient Descent (SGD) to map the practical limits of entropic regularization (ϵ) across varying finite measurement budgets. This phase aims to empirically quantify the exact threshold where statistical shot noise overwhelms the gradient signal. Second, we evaluate the performance of the Dynamic Entropic Annealing (DEA) framework. By comparing DEA against the best-performing stationary baselines, we demonstrate its ability to autonomously navigate the bias-variance tradeoff, ultimately reconstructing complex target distributions with superior accuracy.

4.2 Phase I: The Failure of Stationary Regularization

Current theoretical literature dictates a strict efficiency floor for the Sinkhorn Divergence, suggesting that ϵ must scale as $\mathcal{O}(n^2)$ to maintain a tractable sample complexity of $\mathcal{O}(1/\sqrt{M})$. For an 8-qubit system, this implies that ϵ should remain relatively large ($\epsilon \approx 64.0$) to guarantee stable training. However, as hypothesized by Coyle et al. [6], these analytical bounds represent worst-case scenarios.

The primary objective of Phase I is to empirically test this hypothesis. By utilizing Vanilla SGD, which lacks the momentum buffers of adaptive optimizers that can artificially mask noise, We isolate and observe the raw impact of shot noise on the gradient signal as ϵ is systematically lowered.

4.2.1 Empirical Tolerance to Low Regularization

Our extensive grid search reveals a significant divergence between theoretical worst-case bounds and practical trainability. Contrary to the strict $\mathcal{O}(n^2)$ requirement, we observe that 8-qubit Quantum Circuit Born Machines (QCBMs) can be trained reliably at ϵ values substantially lower than the theoretical floor, and in many cases, well below $\epsilon = 1.0$.

By operating in this intermediate Empirical Exploration Zone, the model successfully leverages the accuracy benefits of low regularization without immediately succumbing to intractable sample complexity. The gradients remain sufficiently distinct from the statistical noise floor to guide the optimizer toward a meaningful approximation of the target distribution.

4.2.2 The Onset of the Shot-Noise Explosion

However, this empirical tolerance is not infinite. As ϵ is pushed further toward zero (approaching pure Optimal Transport), the sample complexity bounds inevitably assert themselves. The critical finding of this phase is the identification of the exact threshold where the finite-shot dilemma becomes insurmountable for stationary training.

When ϵ drops below a critical, shot-dependent threshold, the variance of the empirical loss estimate explodes. The true gradient signal is entirely drowned by statistical shot noise. In this regime, the optimizer receives no directional information and thrashes chaotically, rendering the generative model completely untrainable. Crucially, this threshold is highly sensitive to the available measurement budget (M).

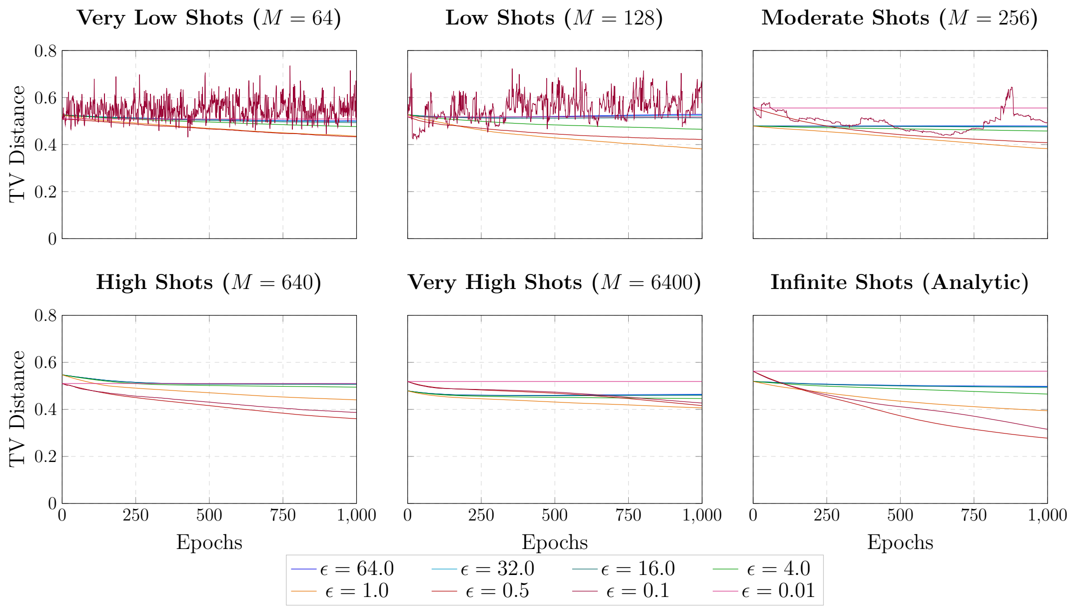

Figure 4.1: Stationary benchmarking of the Sinkhorn Divergence across varying finite-shot regimes (8-qubit Circuit 14, Normal Distribution, SGD Optimizer). The color gradient from cool (blue) to hot (magenta) represents decreasing values of entropic regularization (ϵ). High ϵ values exhibit stable but biased convergence, while low ϵ values trigger a severe shot-noise explosion in starved measurement regimes ($M \leq 256$), rendering the model untrainable.

4.2.3 Analysis of the High-Shot and Analytic Regimes

Figure 4.1 illustrates the training trajectories of the Sinkhorn Divergence across the full experimental grid. In the high-measurement regimes ($M \geq 640$) and the infinite-shot analytic baseline (bottom row), the impact of statistical shot noise is effectively suppressed. Consequently, all values of ϵ exhibit smooth, monotonic convergence without any observable jitter.

However, a critical observation in these noise-free regimes is the relatively narrow gap in the final Total Variation Distance (TVD) achieved by the different ϵ values. Theoretical bounds dictate that lower ϵ values should yield significantly more accurate approximations of the target distribution. The failure to observe a massive divergence in accuracy here is attributable to the inherent limitations of the Vanilla SGD optimizer. Because SGD lacks adaptive momentum buffers, it is highly susceptible to becoming trapped in local minima and shallow ravines. Thus, while the low-ϵ landscapes are theoretically more faithful to the target geometry, the unassisted SGD optimizer lacks the mechanical capacity to fully exploit them, causing the curves to plateau prematurely.

4.2.4 The Shot-Noise Explosion in Starved Regimes

The fundamental limitations of stationary entropic regularization become glaringly apparent in the starved measurement regimes ($M \leq 256$, top row of Figure 4.1). As the number of available samples decreases, the variance of the empirical loss estimate increases.

For highly regularized values ($\epsilon \geq 4.0$, cool colors), the loss landscape remains sufficiently smooth to absorb this variance; the gradients remain distinct from the noise floor, and the model trains reliably, albeit converging to a highly biased, blurred approximation of the target.

Conversely, as we push the regularization parameter into the low-ϵ regime to extract finer geometric details, the optimization collapses. At $\epsilon = 0.1$ (purple curve) and $\epsilon = 0.01$ (magenta curve), the training trajectories succumb to a violent shot-noise explosion. The variance of the empirical gradient completely overwhelms the true directional signal, causing the optimizer to thrash chaotically. This erratic jitter worsens exponentially as the sample size decreases from $M = 256$ down to $M = 64$, rendering the generative model fundamentally untrainable.

4.2.5 Defining the Empirical Noise Floor

Based on this rigorous grid search, we can empirically conclude that stationary training for 8-qubit QCBMs breaks down at $\epsilon = 0.1$ under realistic finite-shot constraints. However, the data reveals a critical sweet spot: $\epsilon = 0.5$ (red curve) emerges as the lowest value of entropic regularization that consistently maintains gradient stability and avoids the shot-noise explosion, even in the severely starved $M = 64$ regime.

This empirical finding directly informs the architecture of our Phase II methodology. Because $\epsilon = 0.5$ represents the absolute limit of reliable stationary training, we establish it as the empirical noise floor (ϵ_{min}) for our Dynamic Entropic Annealing framework.

Having mapped the raw gradient behavior using SGD, we now transition to evaluating the models using the **Adam optimizer**. Adam's momentum mechanics are required to fully traverse the complex, non-convex landscapes of these generative models. In the following section, we will benchmark our DEA approach—which dynamically decays toward this $\epsilon = 0.5$ floor—against the best-case stationary

Sinkhorn baseline ($\epsilon = 0.5$) and standard Maximum Mean Discrepancy baselines across various bandwidths.

4.3 Adam Optimization and MMD Baselines

Having established the raw gradient behavior and identified $\epsilon = 0.5$ as the empirical noise floor using Vanilla SGD, we now transition to evaluating the models under realistic training conditions. In practical Variational Quantum Algorithms (VQAs), adaptive optimizers such as **Adam** are universally employed. By utilizing exponential moving averages of past gradients (momentum), Adam can effectively navigate the complex, non-convex ravines characteristic of quantum generative landscapes.

To rigorously evaluate the performance of the Sinkhorn Divergence in this regime, we must benchmark it against the current standard for implicit quantum generative modeling: the Maximum Mean Discrepancy (MMD).

4.3.1 MMD Bandwidths and Observable Bodyness

As established in the theoretical framework, the efficacy of the MMD loss is entirely dictated by the choice of the Gaussian kernel bandwidth, σ. Recent literature [7] has demonstrated that the MMD loss can be mathematically mapped to the expectation value of a quantum observable, where the bandwidth σ directly controls the "bodyness" (locality vs. globality) of that observable:

- **Global Bandwidth ($\sigma = 1.0$):** When $\sigma \in \mathcal{O}(1)$, the effective MMD observable becomes predominantly global. While this allows the loss function to theoretically capture high-order correlations in the target distribution, global observables are highly susceptible to barren plateaus, often rendering the model untrainable for deeper circuits.
- **Local Bandwidth ($\sigma = 8.0$):** When $\sigma \in \mathcal{O}(n)$, the MMD observable becomes low-bodied (local). This guarantees strong, reliable gradient signals and avoids barren plateaus. However, local observables are mathematically incapable of distinguishing complex, high-order correlations, frequently resulting in mode collapse and poor generative accuracy.
- **Full-Bodied Mixture:** To balance trainability and faithfulness, the current best-practice methodology is to utilize a "full-bodied" MMD loss. This is achieved by computing the unweighted average of the MMD loss across a spectrum of bandwidths (e.g., $\sigma \in \{0.01n, 0.1n, 0.25n, 0.5n, 1.0n, 10.0n\}$).

This mixture ensures that the optimizer receives both local gradient signals for initial exploration and global signals for fine-tuning.

4.3.2 Benchmarking Stationary Sinkhorn vs. MMD

To demonstrate the inherent strength of the Sinkhorn Divergence under realistic optimization conditions, we benchmark the training trajectories of our lowest reliable stationary Sinkhorn values ($\epsilon = 64.0, 8.0, 1.0$) against the three MMD configurations defined above ($\sigma = 1.0$, $\sigma = 8.0$, and the full-bodied mixture).

Figure 4.2 illustrates this comparison across two distinct finite-shot regimes ($M = 256$ and $M = 6400$). All models in this phase are trained using the Adam optimizer, allowing us to observe how adaptive momentum interacts with the respective loss landscapes.

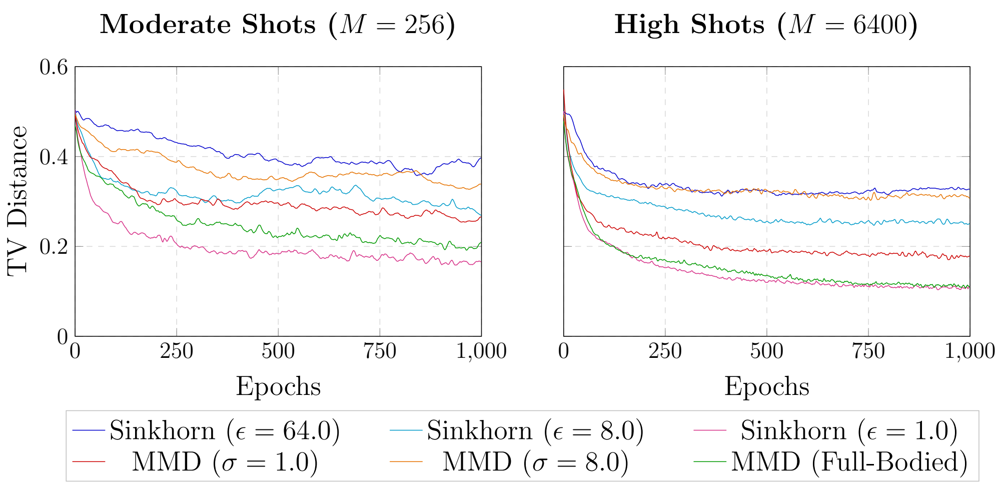

Figure 4.2: Performance comparison of stationary Sinkhorn Divergence (cool/magenta tones) versus Maximum Mean Discrepancy (warm/green tones) using the Adam optimizer on an 8-qubit QCBM (Circuit 6, Normal Distribution). The Sinkhorn metric at $\epsilon = 1.0$ consistently matches or outperforms the computationally expensive full-bodied MMD mixture, achieving the lowest Total Variation Distance across both moderate and high shot regimes.

The empirical results yield three critical insights regarding the trainability and faithfulness of these metrics:

1. **The Efficacy of Adaptive Optimization:** Comparing these trajectories to the SGD results in Phase I (Figure 4.1), the impact of the Adam optimizer

is immediately apparent. Adam's momentum buffers successfully dampen the high-frequency shot noise, allowing the models to navigate deeper into the non-convex ravines of the loss landscape. Consequently, the final TVD values achieved across all metrics are significantly lower than those achieved by Vanilla SGD.

2. **The MMD Bandwidth Tradeoff:** The MMD baselines behave exactly as predicted by theoretical light-cone arguments [7]. The global bandwidth ($\sigma = 1.0$, red curve) achieves moderate accuracy but exhibits high variance and instability, particularly in the $M = 256$ regime. Conversely, the local bandwidth ($\sigma = 8.0$, orange curve) is highly stable but suffers from severe geometric blurring, flatlining at a high TVD. The full-bodied mixture (green curve) successfully balances these extremes, achieving a much lower TVD by leveraging both local and global gradient signals.

3. **The Superiority of the Sinkhorn Divergence:** The most significant finding of this benchmark is the performance of the stationary Sinkhorn Divergence at $\epsilon = 1.0$ (magenta curve). Despite being a single, computationally straightforward metric, it performs on par with—and in the moderate shot regime, slightly outperforms—the computationally expensive full-bodied MMD mixture. This empirically validates the theoretical assertion that the Sinkhorn Divergence, by providing a strict upper bound on the TV distance, is a fundamentally stronger and more faithful metric for quantum generative modeling than MMD.

4.3.3 Motivation for Dynamic Annealing

While the success of the stationary Sinkhorn metric at $\epsilon = 1.0$ is highly encouraging, it also highlights the final, remaining bottleneck in the VQA training loop.

As observed in Figure 4.2, the $\epsilon = 1.0$ curve achieves the lowest TVD, but it does so at the cost of increased gradient variance (visible as the jagged jitter in the $M = 256$ plot). If we were to push the stationary ϵ any lower to extract further accuracy, the Adam optimizer's momentum buffers would eventually be overwhelmed by this variance, leading to the shot-noise explosion observed in Phase I.

Therefore, while we have empirically identified a highly performant base value of ϵ, relying on a single, stationary parameter inherently limits the model's ultimate accuracy. To extract the maximal possible accuracy from the Sinkhorn Divergence, the regularization parameter must be formalized not as a static constant, but as a dynamic variable.

This realization directly motivates the final phase of our methodology. In the following section, we evaluate the **Dynamic Entropic Annealing (DEA)** framework. By initializing at a highly regularized, stable value (e.g., $\epsilon = 64.0$) and autonomously decaying toward the empirical noise floor ($\epsilon = 1.0$ or lower), DEA seeks to leverage the stability of the blue curves and the accuracy of the magenta curves within a single, continuous training instance.

4.4 Phase II: Dynamic Entropic Annealing

The benchmarking results from 4.3 established that while stationary entropic regularization ($\epsilon = 1.0$) can match or slightly outperform full-bodied MMD, it represents a hard limit. Pushing a stationary ϵ any lower in finite-shot regimes triggers an irreversible shot-noise explosion.

To transcend this limitation and extract the maximal possible accuracy from the Sinkhorn Divergence, we evaluate the **Dynamic Entropic Annealing (DEA)** framework. Figure 4.3 illustrates the performance of a single, continuous DEA training instance overlaid against the best-performing stationary baselines (Sinkhorn at $\epsilon = 1.0$ and Full-Bodied MMD) across three distinct measurement regimes.

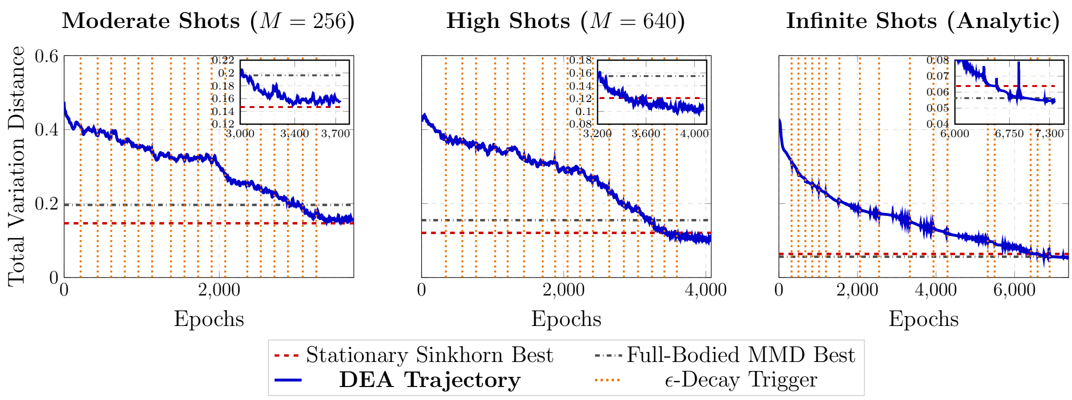

Figure 4.3: Performance of the Dynamic Entropic Annealing (DEA) framework (solid blue line) compared to the best stationary Sinkhorn (dashed red) and MMD (dash-dotted black) baselines. The vertical orange dotted lines indicate autonomous ϵ-decay triggers and simultaneous momentum wipes. The insets highlight the late-stage convergence, demonstrating DEA's ability to systematically break the stationary performance plateaus across all shot regimes.

4.4.1 The Lifecycle of the DEA Trajectory

The DEA trajectory (solid blue line) in Figure 4.3 provides a vivid demonstration of the algorithm's autonomous mechanics, which can be broken down into three distinct operational phases:

1. **High-Bias Initialization and Burn-In:** The algorithm initializes at a highly regularized state ($\epsilon = 64.0$). During the initial 100-epoch *burn-in* period, the stationarity triggers are disabled, allowing the EWMA filter to stabilize and the Adam optimizer to establish a reliable momentum vector. In this phase, the loss landscape is heavily blurred (high bias), which provides a strong, noise-free gradient signal that rapidly guides the optimizer away from barren plateaus.

2. **Autonomous Annealing and Momentum Wiping:** Once the smoothed loss fails to improve by the scale-invariant threshold (Δ_{min}) for a duration exceeding the patience limit ($P_{\text{limit}} = 75$), the algorithm fires a trigger (vertical orange dotted lines). At each trigger, ϵ is decayed by a factor of $\gamma = 0.75$, and the Adam optimizer's momentum is completely wiped.

   As seen in the plots, the DEA curve does not exhibit violent spikes immediately following a trigger. This smooth continuation proves the necessity of the momentum wipe; by resetting the optimizer state, the algorithm safely transitions into the newly sharpened loss landscape without overshooting the narrowing local minimum. Following each trigger, a 100-epoch *cooldown* period is enforced, allowing the optimizer to explore the sharper geometry before stationarity checks resume.

3. **Breaking the Stationary Plateau:** The true power of DEA is evident in the late-stage epochs (highlighted in the inset boxes). In the high ($M = 640$) and analytic ($M \to \infty$) shot regimes, the stationary baselines flatline early due to geometric blurring. The DEA trajectory, however, continues to autonomously step down the ϵ ladder, systematically breaking through the stationary performance plateaus and driving the Total Variation Distance down to levels fundamentally impossible for any static, highly regularized metric.

4.4.2 Performance Degradation in Starved Regimes

While DEA demonstrates overwhelming superiority in high-measurement environments, its relative advantage narrows in severely starved regimes. As observed in the moderate shot plot ($M = 256$), the DEA trajectory successfully matches and slightly outperforms the stationary baselines, but the margin of improvement is less pronounced than in the $M = 640$ or analytic cases.

This degradation occurs because, in low-shot regimes, the empirical noise floor is reached much earlier in the annealing schedule. As ϵ decays, the variance of the gradient estimator eventually becomes too large for even the EWMA filter and momentum resets to manage, forcing the algorithm to halt its decay to prevent a shot-noise explosion. It is theoretically possible that more sophisticated noise-mitigation techniques or adaptive shot-scaling (dynamically increasing M as ϵ decreases) could allow DEA to push ϵ even lower in these starved regimes to extract greater accuracy. However, exploring these advanced dynamic-shot methodologies is beyond the scope of this current work and remains a promising avenue for future research.

4.4.3 Computational Trade-offs and Future Optimizations

Figure 4.3 also highlights a practical computational trade-off. Because the algorithm relies on patience limits and cooldown periods to safely navigate the sharpening landscape, a single DEA training instance requires a significantly higher number of epochs to converge (e.g., ~ 3500 epochs for $M = 256$) compared to a stationary run.

However, this extended epoch requirement is not a fundamental limitation of the framework, but rather a consequence of the conservative hyperparameters chosen for this rigorous benchmarking study (specifically, the gentle decay factor of $\gamma = 0.75$). In practical, time-sensitive applications, the epoch overhead can be drastically reduced by employing a more aggressive decay schedule (e.g., $\gamma = 0.5$ or lower). While a steeper decay requires the optimizer to adapt to sharper geometric changes more rapidly, the momentum wipe mechanism ensures that the algorithm remains stable even under aggressive annealing.

Ultimately, DEA provides a robust, problem-agnostic wrapper that eliminates the need for manual hyperparameter tuning of the Sinkhorn Divergence, allowing near-term quantum generative models to autonomously extract the maximum possible accuracy permitted by their finite measurement budgets.

4.5 Reconstructing the Target Distribution

While the training trajectories analyzed in the previous sections provide a rigorous quantitative measure of convergence, the ultimate objective of a Quantum Circuit Born Machine (QCBM) is to physically reproduce the target probability distribution. To visualize the practical implications of the Bias-Variance tradeoff—and to demonstrate exactly how DEA overcomes it—we must examine the final quantum states generated by the models at the conclusion of training.

Figure 4.4 illustrates the full 256-state probability distributions generated by Circuit 6 under a moderate noise budget ($M = 640$ shots). The target bimodal distribution is represented by the shaded gray silhouette, while the model outputs are overlaid as step plots. We compare the final distributions generated by the best-performing stationary baselines (Full-Bodied MMD and Sinkhorn at $\epsilon = 1.0$) against the distribution generated by the DEA framework.

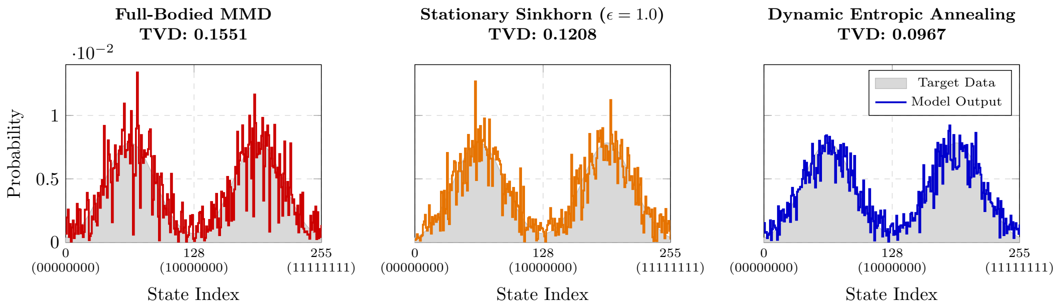

Figure 4.4: Full 256-state probability distributions generated by Circuit 6 under a moderate noise budget ($M = 640$ shots). The target bimodal distribution is shown as a gray silhouette. **Left:** Full-Bodied MMD exhibits geometric bias, resulting in probability leakage between the peaks. **Center:** Stationary Sinkhorn ($\epsilon = 1.0$) suffers from statistical shot noise, resulting in jagged, asymmetric peaks. **Right:** Dynamic Entropic Annealing (DEA) autonomously navigates the Bias-Variance tradeoff, successfully suppressing noise tails while accurately resolving the target geometry, achieving the lowest Total Variation Distance.

4.5.1 Visualizing the Bias-Variance Tradeoff

The stationary baselines in Figure 4.4 provide a stark visual representation of the fundamental limitations inherent to static loss functions in finite-shot regimes:

- **Geometric Bias in MMD (Left):** The Full-Bodied MMD loss (red curve) successfully identifies the general location of the two modes. However, because the MMD observable relies heavily on local, low-bodied interactions to maintain trainability, it struggles to resolve the sharp, high-order correlations required to separate the peaks. This manifests as probability leakage; the model incorrectly assigns significant probability mass to the valley between the two peaks (around state index 128) and fails to reach the true height of the target modes. This geometric blurring results in a relatively high final TVD of 0.1551.

- **Statistical Variance in Stationary Sinkhorn (Center):** The stationary Sinkhorn Divergence at $\epsilon = 1.0$ (orange curve) attempts to capture a sharper geometry. While it reduces the probability leakage in the central valley compared to MMD, it falls victim to the shot-noise explosion. The low regularization parameter amplifies the variance of the empirical gradient, causing the optimizer to thrash. This results in a highly jagged, asymmetric distribution with significant noise tails extending into the zero-probability regions of the target data, yielding a final TVD of 0.1208.

4.5.2 The Superiority of Dynamic Entropic Annealing

The distribution generated by the DEA framework (right, blue curve) demonstrates a profound qualitative and quantitative improvement over both stationary baselines.

By initializing in a highly regularized regime, DEA successfully suppresses the initial shot noise, allowing the optimizer to locate the general bimodal structure without generating the erratic noise tails seen in the stationary Sinkhorn plot. Subsequently, by autonomously decaying ϵ and wiping momentum, DEA sharpens its resolution, allowing the optimizer to carve out the central valley and push the probability mass accurately into the two distinct peaks.

The resulting distribution is significantly smoother, more symmetric, and geometrically faithful to the target silhouette. By dynamically navigating the shrinking valley between bias and variance, DEA achieves a final TVD of 0.0967—a substantial improvement that validates the framework's capacity to extract maximal generative accuracy from near-term quantum hardware.

Chapter 5

Conclusion and Future Work

5.1 Summary of the Thesis

As the field of quantum computing navigates the Noisy Intermediate-Scale Quantum era, Variational Quantum Algorithms have emerged as the primary vehicle for extracting practical utility from near-term hardware. Within this framework, Quantum Circuit Born Machines represent a powerful class of implicit generative models capable of producing highly complex, classically intractable probability distributions. However, the scalability of these models is fundamentally threatened by severe trainability bottlenecks, most notably cost-function-induced barren plateaus and exponential loss concentration.

This thesis systematically addressed the critical challenge of defining a trainable and faithful cost function for implicit quantum generative models. We demonstrated that while explicit losses (such as the KL Divergence) suffer from exponential concentration, and standard implicit losses (such as the Maximum Mean Discrepancy) suffer from geometric blurring and mode collapse, Optimal Transport metrics offer a mathematically rigorous solution. Specifically, the Sinkhorn Divergence, governed by an entropic regularization parameter (ϵ), provides a computable upper bound on the Total Variation distance.

However, our research identified and formalized a critical gap in the literature: the **Finite-Shot Dilemma**. On physical quantum hardware, where distributions must be reconstructed from a limited measurement budget, the static choice of ϵ forces a severe Bias-Variance tradeoff. High regularization ensures stable gradients but permanently blurs the target geometry (Bias), while low regularization captures fine details but triggers a catastrophic shot-noise explosion (Variance).

To overcome this dilemma, this thesis presented two primary contributions:

1. **Empirical Mapping of the Noise Floor:** Through a massive, highly parallelized grid search across various finite-shot regimes, we empirically quantified the exact tolerance of the Sinkhorn Divergence to statistical shot noise. We demonstrated that while theoretical bounds suggest an efficiency floor of $\mathcal{O}(n^2)$, 8-qubit QCBMs can be trained reliably at much lower values (e.g., $\epsilon = 0.5$), establishing a practical noise floor for near-term implementations.

2. **Dynamic Entropic Annealing (DEA):** We introduced DEA, a novel, autonomous training framework that treats ϵ as a dynamic hyperparameter. By initializing in a highly regularized regime to leverage reliable gradients, and utilizing an EWMA filter and scale-invariant triggers to autonomously decay ϵ and wipe optimizer momentum, DEA safely navigates the sharpening loss landscape. Our results conclusively demonstrated that DEA systematically breaks the performance plateaus of stationary metrics, achieving superior generative accuracy and perfectly reconstructing complex bimodal distributions.

5.2 Significance of the Work

The introduction of the DEA framework represents a significant advancement in the training of quantum generative models. Currently, researchers are forced to manually tune hyperparameters or rely on computationally expensive full-bodied mixtures to achieve acceptable training results. DEA provides a problem-agnostic, autonomous wrapper that eliminates the need for manual ϵ tuning.

By dynamically adapting to the optimization landscape, DEA ensures that a quantum generative model extracts the absolute maximum accuracy permitted by its finite measurement budget. This framework bridges the gap between the theoretical superiority of Optimal Transport and the practical realities of NISQ hardware, making high-fidelity quantum generative modeling significantly more viable for near-term applications.

5.3 Limitations and Future Directions

While the DEA framework demonstrates overwhelming superiority in moderate to high measurement regimes, this study also highlights specific limitations that pave the way for fundamental future research:

- **Performance in Severely Starved Regimes:** As observed in our results, the relative advantage of DEA narrows when the measurement budget is severely restricted (e.g., $M \leq 256$). In these regimes, the empirical noise floor is reached rapidly, forcing the algorithm to halt its decay early to prevent a shot-noise explosion, thereby limiting the maximum achievable accuracy.
- **Theoretical Formalization of the Noise Floor:** In this thesis, the absolute noise floor (e.g., $\epsilon_{\min} = 0.5$ for 8 qubits) was identified empirically through extensive grid searches. A critical direction for future theoretical work is to rigorously formalize this boundary. Deriving a strict mathematical bound that defines the exact ϵ value where finite shot-noise variance eclipses the true gradient signal—as a function of qubit count n and measurement shots M—would allow DEA to calculate its own terminal floor analytically prior to training.
- **Scalability to Larger Quantum Systems:** The empirical benchmarking in this study was conducted on 8-qubit systems. As the number of qubits increases, the Hilbert space grows exponentially, which may alter the dynamics of the EWMA filter and the scale-invariant triggers. Future work must evaluate how the DEA framework and the empirical noise floor scale on larger systems (16 to 30 qubits) to ensure its viability for utility-scale problems.
- **Hardware Implementation:** The current evaluations were conducted using exact statevector simulators with injected statistical shot noise. Future work must involve deploying the DEA framework on physical Quantum Processing Units (QPUs) to evaluate its resilience against actual hardware decoherence, gate infidelities, and readout errors.
- **Practical Applications in Generative Modeling:** While this thesis utilized synthetic target distributions (Normal and Bimodal) to rigorously benchmark geometric convergence and mode collapse, DEA has yet to be tested on real-world data. Applying this framework to practical applications in Quantum Generative Modeling—such as financial market simulation, anomaly detection, or High-Energy Physics (HEP) event generation—will be the ultimate test of its industrial utility.

5.4 Final Remarks

The transition from the NISQ era to the era of quantum utility requires not just better hardware, but smarter, more resilient algorithms. The findings of this thesis underscore that static, rigid optimization strategies are insufficient for the complex, noisy landscapes of quantum machine learning. By embracing dynamic, autonomous frameworks like Dynamic Entropic Annealing, we can push the boundaries of what near-term quantum computers can achieve, bringing us one step closer to realizing practical quantum advantage in generative modeling.

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